Home Advantage in Japan's Kokutai: Evidence from 77 Editions and Sport-Type Heterogeneity — Non-Monotonic Time Pattern and Subjective-Objective Sport Divide

Running title: Home advantage across 77 Kokutai editions

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Abstract

Background: Host-country advantage in international sport is well established across the Summer and Winter Olympic series (Balmer, Nevill & Williams 2001, 2003; Wilson & Ramchandani 2021), and Balmer, Nevill & Williams (2003) further proposed that home advantage should be concentrated in *subjectively judged* events (boxing, gymnastics) rather than in objectively measured events (track, weightlifting). Japan's National Sports Festival (国民体育大会, *Kokutai*; renamed 国民スポーツ大会 in 2024) is an unusually long-running natural experiment for these questions — 47 prefectures rotating as hosts under essentially identical eligibility rules since 1946 — yet the leading Japanese-language predecessor (Funahashi et al. 2016) analyzed only a 9-year window (2003-2011) and did not test either the full time series for the reality of the host bonus or the objective/subjective decomposition. Using the full editions 3-79 (1948-2025), we ask three primary questions: (i) does the host prefecture reach top-1, top-3, and top-8 at rates significantly exceeding the chance baseline of $k/47$? (ii) is the host bonus stable across the 75 completed editions in the 3-79 span (77 editions total, of which 2 were cancelled for COVID-19), or does it vary by era? (iii) is the host bonus larger in subjectively judged sports than in objectively measured ones?

Methods: We assembled a host-prefecture ranking panel spanning the 77 editions from the 3rd (1948) through the 79th (2025) meet, of which 75 were non-cancelled and included in the analysis (n = 150 host-observations = 75 non-cancelled meets × 2 trophies; the emperor's cup for combined male-female standings and the empress's cup for female-only; the 75th (2020) and 76th (2021) editions were cancelled for COVID-19), parsed from the Nagano Prefecture Sports Association ranked-listings page (independently cross-validated against JSPO official archive PDFs for the 58th-67th editions: 20 of 20 host-year cells matched exactly). Editions 1-2 (1946-1947, no complete official ranking data) and the 75th-76th editions (2020-2021, cancelled for COVID-19) were excluded. The primary analyses were (i) a

one-sample proportion test against the chance null of $k/47$ (exact binomial with two-sided Wilson 95% confidence intervals; $k \in \{1, 3, 8\}$), (ii) a permutation test in which host-prefecture identity was re-drawn uniformly at random from the 47 prefectures (n_perm = 10,000, Monte Carlo p-value lower-bound correction (count + 1) / (n_perm + 1) following Phipson & Smyth (2010)), (iii) a three-era χ^2 test with pairwise Fisher exact contrasts partitioning the 77 editions into an early formative era (1948-1977, n = 30), a golden host-victory era (1978-2015, n = 38), and a post-2016 shock era (2016-2025, n = 7), and (iv) a pooled ordered logit of ordinal host rank on era and trophy indicators with cluster-robust standard errors at the host-prefecture level. As a subsidiary analysis, we tested the Balmer et al. (2003) subjective-versus-objective decomposition on a 47-prefecture × 40-sport stacked cross-section of the 2024 Saga and 2025 Shiga editions (n = 6,991 cells; 47 clusters, 2 treated). As a robustness check for the pre-1972 pre-Okinawa-reversion 46-prefecture regime, we recomputed the one-sample tests on the early era under a conservative worst-case 46-state null.

Results: Across all 75 non-cancelled meets, the host prefecture reached the championship in 57 of 75 emperor's cup editions (76.0%; Wilson 95% CI [0.652, 0.842]) and 47 of 75 empress's cup editions (62.7%; [0.514, 0.727]) — respectively 36× and 29× the chance rate of 1/47 = 2.13% (both $p < 10^{-50}$, exact binomial; safe upper cap — the true tail probabilities are far smaller, but a shared cap is reported here in preference to hyper-precise exponents). At the top-3 threshold, the host reached the top 3 in 69 of 75 emperor's cup editions (92.0%) and 64 of 75 empress's cup editions (85.3%), against a chance rate of 3/47 = 6.38%. At the top-8 threshold, both cups reached 93.3% (70/75), against a chance rate of 8/47 = 17.02%. The permutation test corroborated these results with all six threshold-by-cup combinations yielding p = 1/10,001 (the Monte Carlo lower bound). The three-era χ^2 tests, however, revealed a **non-monotonic** time pattern: for the pooled top-1 rate across both cups, the early era produced 43.3%, the golden era 94.7%, and the shock era 42.9% ($\chi^2 = 46.76$, df = 2, $p < 10^{-10}$; pairwise Fisher exact: early vs. golden $p < 10^{-10}$, golden vs. shock $p < 10^{-5}$, early vs. shock p = 1.00) — the two endpoints indistinguishable and only the golden era differing sharply from both. The pooled ordered logit confirmed this shape with era_golden coef = -3.33 (SE = 0.86, p < 0.001; negative = better rank) and era_shock coef = -0.54 (SE = 0.54, p = 0.317) relative to the early era baseline. The 46-versus-47-state sensitivity check found that the excess above chance differs by less than 0.005 across all early-era threshold-cup cells, so the default 47-state null mildly overstates excess but the impact is marginal. In the 2024-2025 sport-level cross-section, the host × subjective interaction was +16.68 points (cluster-robust SE = 7.72, p = 0.031); the wild-cluster bootstrap (Rademacher weights, restricted null, B = 999) yielded p = 0.155 under the few-treated-clusters correction (only 2 of 47 clusters treated) and should be read as directional rather than decisive. The descriptive host-boost gap ran monotonically from +37.9 points in subjective sports through +26.2 in semi-subjective/team to +17.6 in objective, in the direction Balmer et al. (2003) predicts.

Conclusions: The reality of the Kokutai host bonus is unambiguous when tested against the $k/47$ chance baseline across the 75 completed editions (editions 3-79 span; 2 COVID cancellations excluded): the host reaches the top-8 in 93.3% of editions against a 17.02% chance rate, and the championship in 76.0% of emperor's-cup editions against a 2.13% chance rate. But the effect is **non-monotonic** across eras — the early formative period and the post-2016 shock period are statistically indistinguishable in host-victory

rate, and only the 1978-2015 golden era stands sharply apart. A stable single-mechanism explanation (host quota, facility familiarity, or crowd support) does not fit this shape; the pattern is more consistent with a structural feature specific to 1978-2015. The Balmer et al. (2003) subjective-versus-objective decomposition is directionally supported in the 2024-2025 sport-level cross-section (+16.68 interaction; monotonic +37.9/+26.2/+17.6 descriptive gradient) but does not survive the few-treated-clusters correction (bootstrap $p = 0.155$). The two central findings — the 75-completed-edition reality test and the non-monotonic time pattern — are the main contribution; the sport-level Balmer decomposition is a directionally consistent first empirical anchor pending broader sport-level historical data (which the JSPO archive does not currently release for editions prior to 2024). Historical sport-level records for editions 1-65 exist in the JSPO's 65-edition compendium but their tabular structure (three-column layout, top-6 finishing places with frequent multi-way ties, mixed institutional and prefectural affiliations for individual events) precludes reliable structured parsing at present, and their integration is left to future work.

Keywords: Home advantage, host effect, National Sports Festival, Kokutai, one-sample proportion test, permutation inference, non-monotonic time pattern, subjective judging, Balmer 2003, Japan

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Introduction

Does hosting a sporting event genuinely improve the host's performance, and if so, through what mechanism? At the international scale, the answer is unambiguously yes: host countries reliably win more medals than their non-host baseline predicts, an effect documented across a century of Summer Olympic Games (Balmer, Nevill & Williams 2003)[3] and the Winter Olympic series (Balmer, Nevill & Williams 2001)[1], with a broader Olympic/Paralympic replication (Wilson & Ramchandani 2021)[2] over the modern 1988-2018 window. The mechanism, however, is contested. Six candidate pathways are typically invoked — host-country entry quotas, training-venue familiarity, home-crowd support, budget uplift, jet-lag/climate advantage, and, most contentiously, **subjective-judging bias in favor of the host** — but no single mechanism has been uniquely identified in any single dataset.

Balmer, Nevill & Williams (2003) proposed a diagnostic decomposition.[3] If home advantage were driven by facility familiarity, budget, or crowd support, it should appear roughly uniformly across all Olympic sports. If, on the other hand, referee or judge bias were a load-bearing mechanism, home advantage should be concentrated in sports where the outcome depends on human scoring rather than an objective clock or tape measure. Comparing five event families across a century of summer Olympic data, Balmer et al. (2003) found that home advantage is indeed disproportionately concentrated in subjectively judged sports — boxing and gymnastics, in the specific event groups reported in the paper's abstract — with much smaller or absent effects in objectively measured sports such as athletics and weightlifting. This decomposition provides an implicit test of the judging-bias mechanism that does not require access to individual judge scores.

Japan's National Sports Festival (国民体育大会, *Kokutai*; renamed 国民スポーツ大会 in 2024) offers an unusually long-running setting for testing whether these results generalize. Since 1946, the Kokutai has rotated across all 47 prefectures on a fixed schedule, with essentially identical eligibility rules and a stable overall-score system. The event is large (approximately 40 sports across a summer meet and a smaller winter meet), long-running (79 editions completed through 2025 with an 80th scheduled for autumn 2026 in Aomori), and centrally administered by the Japan Sport Association (JSPO). Three headline patterns are widely known within Japan. First, host prefectures win at a startling rate: across the full 1948-2025 series (editions 3-79, excluding the never-completely-ranked 1946-1947 first two editions and the 2020-2021 COVID-cancelled meets), the host prefecture reaches the emperor's cup championship in 57 of 75 non-cancelled meets (76.0%). Second, this aggregate figure masks a striking non-monotonic time pattern: from 1978 through 2015 (the "*golden era*" of host wins in the local parlance), the host prefecture took the emperor's cup championship in **37 of 38 editions (97.4%)**, whereas in the post-2016 shock era the host championship rate has collapsed to **3 of 7 (42.9%)** across the 7 non-cancelled meets in 2016-2025 (the 2020 Kagoshima and 2021 Mie editions were cancelled for COVID-19); the shock era's championship rate is thus statistically indistinguishable from that of the pre-1978 early era (26 of 60 pooled cups = 43.3%; pairwise Fisher exact p = 1.00), and both differ sharply from the golden era. Third, the post-2016 collapse has generated both journalistic speculation and one scholarly critique (Suetsugu, 2024, 2025)[4,5] that host outcomes reflect a "host-victory-first mindset" (a phrase corresponding to the "開催地優勝至上主義" framing that appears in the title of Suetsugu 2025[5]). (For the historical record, the popular phrase "*Nara hantei*" — a slur for perceived hometown scoring bias — refers to a 2018 boxing scandal centered on Yamane Akira, then president of the Japan Amateur Boxing Federation, as documented in Japanese national news coverage of the federation's July 2018 self-dissolution and subsequent JOC/JSPO governance reforms; the phrase does not refer to the Kokutai kendo competition as is sometimes claimed.)

The most substantial quantitative predecessor is Funahashi, Hibino, Ishiguro & Mano (2016).[6] They assembled a balanced 47-prefecture × 9-year panel (n = 423) for 2003-2011, regressed the male-female combined competition score on a full set of host-year event-time dummies (t-7 through t+7), and estimated a host-year bonus of +1,674.65 total-score points on top of prefecture and year fixed effects (R^2 = 0.86). Funahashi (2016)'s Table 4 lists the six candidate mechanisms invoked above and cites Balmer et al. (2001)[1] — Balmer's Winter Olympics paper — as their reference for the subjective-judging pathway. But their analysis is bounded in three separate ways relative to what a full assessment of the Kokutai host effect requires. First, they **did not extend the analysis window beyond 2011**, and consequently could not test whether the host bonus is stable across eras — the post-2016 collapse noted above lies entirely outside their sample. Second, they **did not test the reality of the host bonus against a formal chance baseline** (such as the $k/47$ uniform-selection null against which host performance in the top-k is naturally benchmarked); their fixed-effects regression documents that host years score higher than non-host years for the same prefecture but does not directly test whether the host prefecture's top-k finishing rate exceeds what would occur under random assignment of the host role. Third, they **did not cite Balmer et al. (2003) or include either a sport-type indicator or a host × sport-type interaction** in any of their specifications, and the subjective-versus-objective decomposition that Balmer et al. (2003) formalized is therefore absent from the Japanese panel literature. The 75-completed-edition reality test (span editions 3-79), the era-heterogeneity analysis, and the Balmer et al. (2003) subjective-versus-objective decomposition are the three specific gaps this paper addresses. Chiba's earlier qualitative treatment (1987)[7] likewise enumerates five host-victory mechanisms — the full-entry system, alleged combination-drawing bias, transferred-athlete contributions, intra-prefecture athlete development, and prefecture-wide civic support — and explicitly does not mention refereeing or judging bias in any of the five.

Institutionally, the Japanese Kokutai stands in mild contrast to the Korean equivalent. The Korean Sport & Olympic Committee's national comprehensive sports competition regulations explicitly award a **20% bonus** to the host city's overall score, codified from 2010 onward (previously 10% from 2001).[8] Japan has no such explicit host bonus in its scoring rules — the JSPO regulations we examined contain no analogous provision — which makes the Japanese host advantage, if present, a purely emergent phenomenon rather than an institutional artifact. This makes the reality test especially informative in the Japanese case: a host bonus that appears at extreme magnitudes in the absence of an institutional multiplier is more strongly suggestive of the constellation of behavioral mechanisms (host-quota selection, facility familiarity, crowd support, and — potentially — subjective-judging bias) that Balmer et al. (2003) and the subsequent literature have proposed.

We accordingly ask three questions. First (**reality test**), does the host prefecture reach top-k finishing positions at rates significantly exceeding the pure-chance baseline of $k/47$ across the 75 completed editions in the 3-79 span (77 editions total, of which the 75th (2020 Kagoshima) and 76th (2021 Mie) were cancelled for COVID-19), and does this reality survive across the top-1 (championship), top-3, and top-8 thresholds so that the finding is not fragile to a single threshold choice? Second (**era heterogeneity**), is the host bonus stable across these 75 completed editions, or does it vary systematically by era — and if the latter, is the variation monotonic (a smooth trend from 1948 through 2025) or non-monotonic (with the pre-1978 early era and the post-2016 shock era resembling each other more than either resembles the 1978-2015 golden era)? Third (**subjective-versus-objective decomposition**), following Balmer et al. (2003), is the host bonus larger in subjectively judged sports (boxing, judo, gymnastics, kendo, karate, …) than in objectively measured ones (track, swimming, cycling, weightlifting, …) in the 2024 Saga and 2025 Shiga editions — the only two editions for which JSPO publicly releases the full 47-prefecture × ~40-sport disaggregation? Our answers, in brief, are: (i) **yes, unambiguously** — for the emperor's cup the host reaches the championship in 76.0% of the 75 non-cancelled editions versus the chance rate of 2.13% (both cups $p < 10^{-50}$ by one-sample exact binomial; the top-3 rate is 92.0% versus 6.38%, and the top-8 rate is 93.3% versus 17.02%, so the finding is stable across all three thresholds); (ii) **non-monotonic** — the early formative era (43.3% pooled top-1 rate, 1948-1977) and the post-2016 shock era (42.9%, 2016-2025) are statistically indistinguishable (pairwise Fisher exact $p = 1.00$), and only the 1978-2015 golden era at 94.7% differs sharply from both ($\chi^2 = 46.76$, df = 2, $p < 10^{-10}$); and (iii) **directionally consistent with Balmer et al. (2003)** — the descriptive host-boost gap runs monotonically from +37.9 points in subjective sports through +26.2 in semi-subjective/team to +17.6 in objective, and the regression host × subjective interaction is +16.68 points (cluster-robust $p = 0.031$); however, this interaction does not survive the wild-cluster bootstrap correction for the two-treated-prefectures structure (p = 0.155), and should be read as a directionally consistent first empirical anchor rather than a decisive rejection of the null pending broader sport-level historical data.

Methods

Data sources

The primary analysis dataset is a host-prefecture ranking panel of all 75 non-cancelled editions in the span from the 3rd (1948) through the 79th (2025) meet (77 editions in total in this span, of which 2 were cancelled), yielding n = 150 host-observations (75 non-cancelled meets × 2 trophies: the emperor's cup (天皇杯) for combined male-female standings and the empress's cup (皇后杯) for female-only standings). The 1st (1946) and 2nd (1947) editions were excluded because no complete official ranking data are available (see Limitations); the 75th (2020 Kagoshima) and 76th (2021 Mie) editions were excluded because they were cancelled for COVID-19. Host-prefecture ranks were parsed from the Nagano Prefecture Sports Association ranked-listings page (https://www.nagano-sports.or.jp/kokutai/record/high_rank.html), which reports the top 8 finishers per edition per trophy across the full editions 3-79 window. To validate the primary source, we independently cross-checked host-prefecture ranks against JSPO official archive PDFs (https://www.japan-sports.or.jp/kokutai/tabid183.html) for the 58th-67th editions (2003-2012) — the range for which the JSPO machine-readable PDFs are reliably parsable with `pdfplumber` (Python 3.14) — and confirmed that **20 of 20 host-year cells (10 editions × 2 trophies) matched exactly** between the two sources. For editions 68-77 (2013-2022), the corresponding JSPO PDFs were either 404 responses or image-only files from which text could not be extracted; the Nagano source provides host-prefecture ranks for these editions and is used as the sole source in that range. For editions 78-79 (2024 Saga and 2025 Shiga), JSPO also published Excel disaggregations of the full 47-prefecture × ~40-sport breakdown (`78_score_data.xls`, `78_score_all_data.xls`, `79_score_all_data.xls`, `79_score_deta.xls`), parsed with `python-calamine` (the Rust-based `xlrd` replacement, required because both files use a formula function ID that `xlrd` misidentifies); these Excel files are used only for the subsidiary sport-level cross-section (Study design item Subsidiary, below), not for the primary host-rank panel.

Because the 3rd-79th editions include three multi-prefecture co-hosted meets (the 7th in 1952 Fukushima/Miyagi/Yamagata, the 8th in 1953 Ehime/Kagawa/Tokushima/Kochi, and the 48th in 1993 Kagawa/Tokushima), the primary host-rank panel normalizes each co-hosted edition to its single administrative host prefecture (defined by the JSPO official record: Fukushima for the 7th, Ehime for the 8th, and Kagawa for the 48th). For the permutation test only (Study design item 2, below), the co-hosted definition is relaxed to "any co-hosting prefecture in the top *k*", to match the pooled 1-of-47 chance null against which co-hosting inflates the probability of at least one co-host reaching the top. Both definitions yielded the same observed top-1 count on the current sample (57 emperor's-cup and 47 empress's-cup out of 75 non-cancelled meets), so the primary results do not depend on which definition is used; the definitional difference is documented in `src/analysis_main_v3.py:permutation_test`.

For the subsidiary 2024-2025 sport-level cross-section, sports were classified as **objective** (n = 17: track and field, swimming, cycling, rowing, canoe, weightlifting, sailing, ski, skating, archery, kyudo, rifle shooting, clay-target shooting, golf, bowling, triathlon, sport climbing; clay-target shooting was appended in the 2024 edition and removed in 2025, so the effective objective count is 17 in 2024 and 16 in 2025), **subjective** (n = 11: kendo, judo, gymnastics, karate, jukendo, naginata, boxing, fencing, wrestling, sumo, equestrian), or **semi-subjective / team** (n = 13: team ball sports with referee-mediated but primarily objective outcomes: soccer, basketball, volleyball, handball, hockey, rugby, softball, baseball, badminton, table tennis, soft tennis, tennis, ice hockey), following Balmer, Nevill & Williams (2003).[3] The classification is implemented in `src/sport_classifier.py` and unit-tested against the Balmer et al. (2003) decision rules.

Study design

The primary analysis is organized around four specifications applied to the 150 host-observations (75 non-cancelled meets × 2 trophies), complemented by one subsidiary specification on the 2024-2025 sport-level cross-section and one sensitivity check on the pre-1972 46-state regime.

1. One-sample proportion test (primary reality test). For each combination of threshold $k \in \{1, 3, 8\}$ and cup ∈ {emperor's, empress's, pooled both}, we test the null hypothesis that the host prefecture's top-k rate equals $k/47$ (the chance rate under uniform random selection of the host role from the 47 prefectures) against the one-sided alternative that the true rate exceeds $k/47$. Test statistic: the exact binomial tail probability (`scipy.stats.binomtest(..., alternative='greater')`). Confidence intervals: two-sided Wilson 95% CI (`scipy.stats.binomtest(...).proportion_ci(0.95, method='wilson')`), reported directly rather than as one-sided CIs from the same test to avoid the trivial-upper-bound-1.0 artifact that a one-sided binomial CI would produce. Sample sizes per cup: 75 non-cancelled meets in total; 30 in the early era (1948-1977), 38 in the golden era (1978-2015), 7 in the shock era (2016-2025). The specification is deliberately covariate-free: the confirmatory reality test asks only whether the observed host top-k rate exceeds the $k/47$ chance baseline, not whether that exceedance is attributable to any particular set of confounders; introducing covariates such as prefecture size or gross domestic product (GDP) would either strengthen or attenuate the excess numerically but would not alter the reality-test conclusion, and would risk a covariate-choice degree of freedom that the reality test is designed to avoid.

2. Permutation test (statistical robustness). For each threshold-cup combination, host-prefecture identity is re-drawn uniformly at random from the 47 prefectures independently for each edition (n_perm = 10,000; the null draws are i.i.d. per edition, so the permutation null is a bootstrap-of-labels rather than a fixed-margin permutation of the same 75 hosts). The Monte Carlo p-value is computed with the Phipson & Smyth (2010) lower-bound correction: p_perm = (count + 1) / (n_perm + 1), so a value of 1/10001 ≈

10^{-4} is the minimum observable p-value and should be read as "no null draw among 10,000 produced an outcome at least as extreme as observed" rather than as a literal 10^{-4} point estimate of the true tail probability. The two co-hosted meets (7th 1952, 8th 1953) enter the permutation null with a single at-random draw per edition to preserve the 1-of-47 chance baseline, matching the one-sample test's definition of the chance null; the definitional relationship to the one-sample test is documented in Data sources above.

3. Three-era χ^2 test with pairwise Fisher exact contrasts (era heterogeneity). The 77 non-cancelled editions are partitioned into three eras: **early** (1948-1977, kai 3-32, 30 meets), **golden** (1978-2015, kai 33-70, 38 meets), and **shock** (2016-2025, kai 71-79 excluding cancelled 75-76, 7 meets). The 1978 boundary corresponds to the widely-cited onset of the golden era of host wins in the Japanese Kokutai literature; the 2016 boundary corresponds to the first host loss in more than a decade (2016 Iwate). For each threshold-cup combination, the joint null that the three era top-*k* rates are equal is tested with a Pearson χ^2 test on a 3×2 contingency table (`scipy.stats.chi2_contingency`; the Yates continuity correction is inapplicable to 3×2 tables and is not invoked by scipy in this case), and complemented by the three pairwise Fisher exact tests (`scipy.stats.fisher_exact`) so that the non-monotonic vs. monotonic character of any era heterogeneity can be read directly from the pairwise results (e.g., a monotonic time trend would produce three ordered pairwise contrasts, whereas a non-monotonic pattern would produce two significant pairwise contrasts flanking one non-significant contrast between the two endpoint eras). The three-era partition is chosen a priori against the qualitative history of the Kokutai host bonus (the 1978 boundary corresponds to the widely-cited onset of the "golden era" of host wins in the Japanese Kokutai literature, and the 2016 boundary corresponds to the first host loss in more than a decade — 2016 Iwate) rather than data-mined from the top-*k* rates themselves; no formal preregistration record (OSF, AsPredicted, or equivalent timestamped protocol) accompanies the partition, and the partition itself is a researcher degree of freedom that we address in the ordered-logit specification below.

4. Pooled ordered logit (era heterogeneity, regression form). As a regression-based cross-check on the era-heterogeneity result, we fit a pooled ordered logit of ordinal host rank on era and trophy indicators (with fixed effects (FE) for era and cup as detailed below): `rank_ordinal ~ C(era, Treatment('early')) + C(cup, Treatment('kougou'))`, where `rank_ordinal` is the host's finishing rank censored at 9 (a placeholder for "outside top 8"), `era` is a three-level categorical with `early` as the reference, and `cup` is a two-level categorical with the empress's cup as the reference. The specification is fitted in three modes: (a) pooled across both cups (n = 150, cup as a covariate); (b) emperor's cup only (n = 75, cup dropped); (c) empress's cup only (n = 75, cup dropped). Standard errors are clustered at the host prefecture code (`host_pref_code`; the 46 distinct host prefectures represented in the 75-meet series (46 of the 47 Japanese prefectures have hosted at least one Kokutai in the 3rd-79th editions)) to account for repeated hosting by the same prefecture. The ordered logit is fitted with `statsmodels.miscmodels.ordinal_model.OrderedModel`, and a graceful fallback to

Fisher-information (unclustered) SEs is implemented for the rare case in which the cluster meat matrix is singular (not triggered on the current sample; the graceful-fallback code path is not currently unit-tested — see Limitations).

Subsidiary. 2024-2025 sport-level cross-section (Balmer et al. (2003) subjective-versus-objective decomposition). The Balmer et al. (2003)-style host × sport-type interaction is tested on a stacked cross-section of **n = 6,991 prefecture-sport-year cells** from the 2024 Saga and 2025 Shiga editions (47 prefectures × ~40 sports × 2 years × 2 trophies; the theoretical full cartesian product would be 47 × [(40 tennou + 35 kougou)_2024 + (40 tennou + 36 kougou)_2025] = 47 × 151 = 7,097 cells, and the regression frame retains 6,991 after dropping 106 unbalanced cells for prefectures that did not participate in specific sports and are absent from the JSPO overall-standings publication for that sport-year, a 1.5% shortfall). The specification is `score_isc = α + β_H · host_isc + β_S · subj_c + β_HS · (host × subj)_isc + η_i + κ_c + ν_t + ξ_p + ϵ_isc` where `subj_c` is a **binary indicator (1 if sport `c` is subjective, 0 otherwise)** — the 13 semi-subjective/team sports are grouped with the 17 objective sports as the reference category (`subj_c = 0`), so β_H is the host bonus for the combined objective + semi-subjective/team reference group. See Limitations §Seventh for the identifying-variance implications of this binary coding on the 2024-2025 sample. The specification has prefecture (`η_i`), sport (`κ_c`), year (`ν_t`), and trophy (`ξ_p`) fixed effects; prefecture-clustered robust standard errors (47 clusters, Liang-Zeger sandwich). Because only 2 of 47 clusters are treated (Saga in 2024, Shiga in 2025), the cluster-robust p-value is complemented by a wild-cluster bootstrap (Rademacher weights, restricted null, B = 999, fixed seed for reproducibility) as the appropriate few-treated-clusters correction (Cameron, Gelbach & Miller 2008; MacKinnon & Webb 2017). We treat this subsidiary specification as directionally informative rather than as a formal confirmatory test given the two-treated-clusters structure. A log-outcome specification (`log(1 + score)`) is reported as a robustness check on functional form; the same wild-cluster bootstrap is applied.

Sensitivity check. Pre-1972 46-state regime. Okinawa did not rejoin Japan until 1972, so editions 3-26 (1948-1971) had only 46 participating prefectures rather than 47. To bound the effect of this small change in the chance baseline, we re-compute all early-era one-sample proportion tests under a 46-state null (`null_rate = k/46` in place of `k/47`), applied **uniformly across the full 30-edition early era** (a conservative worst-case bound rather than an exact recompute restricted to the 24 pre-1972 editions; the additional 6 editions from 1972-1977 had 47 participating prefectures, so applying the 46-state null to them slightly under-states the chance baseline for those 6 specific years but produces a strict upper envelope for the 46-vs-47 effect-size difference across the full early era). Because `null_rate = k/n_states` is a decreasing function of `n_states`, the default 47-state null yields a slightly larger `excess = obs_rate − null_rate` than the 46-state null; a marginal excess-difference between the two nulls therefore indicates that the default 47-state analysis *mildly overstates* excess (not understates), and is the direction relevant to a Limitations claim of the form “the default 47-state analysis is a slight upper bound on the true early-era excess”.

Statistical software

All analyses were performed with Python 3.14 using pandas 3.0.5, numpy 2.5.1, statsmodels 0.14.6, patsy 1.0.2, scipy 1.18.0, python-calamine 0.8.2, pdfplumber 0.11.10, matplotlib 3.11.1, pyarrow 25.0.0, pytest 9.1.1, and (for the PDF build pipeline) weasyprint 69.0 and markdown 3.10.2. The primary one-sample proportion tests use `scipy.stats.binomtest` with `alternative='greater'` for one-sided p-values and Wilson two-sided 95% CIs; the permutation tests use `numpy.random.default_rng` (fixed seed for reproducibility) with the Phipson & Smyth (2010) Monte Carlo lower-bound correction; the three-era χ^2 tests use `scipy.stats.chi2_contingency` (Yates continuity correction inapplicable to 3 × 2 tables) and `scipy.stats.fisher_exact` for pairwise contrasts; the pooled ordered logit uses `statsmodels.miscmodels.ordinal_model.OrderedModel` with cluster-robust standard errors at the host prefecture code (46 unique clusters — 46 of the 47 Japanese prefectures have hosted at least one Kokutai in the 3rd-79th editions). For the subsidiary 2024-2025 cross-section, prefecture-cluster-robust standard errors (47 clusters, Liang-Zeger sandwich) are reported alongside a wild-cluster bootstrap (Rademacher weights, restricted null, B = 999) as the few-treated-clusters correction. The one-sample proportion tests are one-sided at $\alpha = 0.05$ because the alternative hypothesis (host exceeds chance) is directional by construction; all other statistical tests are two-sided at $\alpha = 0.05$ unless otherwise noted. Unit tests (n = 326 total, all passing; runtime ≈ 190 s on Python 3.14) verify the numerical integrity of each analysis module against small deterministic fixtures.

Multiple comparisons. The confirmatory test of this paper is the one-sample proportion test of the host top-*k* rate against the *k*/47 chance null (Question (i), reality test), which is inherently a family of six tests across the three thresholds $k \in \{1, 3, 8\}$ and two cups (emperor's and empress's). All six test statistics reject the null at Monte Carlo minimum p = 1/10001 under the permutation robustness check, so no p-value adjustment is required for the reality test to survive any conventional multiple-comparisons correction (under a Bonferroni correction with $\alpha\_family = 0.05$ across all six one-sample tests, the required per-test threshold is $\alpha\_single \approx 0.0083$, and all six observed one-sample p-values are far below this threshold). The era heterogeneity test (Question (ii)) is a single joint 3-era χ^2 test per threshold-cup pair, complemented by three pairwise Fisher exact contrasts, and is treated as a single confirmatory test per cup/threshold rather than as a family; the paper's headline era-heterogeneity claim (non-monotonic pattern with early ≈ shock ≪ golden) rests on the pooled-both-cups top-1 result ($\chi^2 = 46.76$, df = 2, $p < 10^{-10}$) and does not depend on p-value adjustment across thresholds. The subsidiary Balmer decomposition (Question (iii)) is a single confirmatory interaction test on the 2024-2025 cross-section; because it fails to reject under the appropriate few-treated-clusters wild-cluster bootstrap (p = 0.155), no multiple-comparisons adjustment for the descriptive within-category boost figures is required.

Ethical considerations

This study analyzes publicly available Kokutai overall-standings data released by the Japan Sport Association and prefectural sport associations. No individual-level, health, or personally identifying data are used, and no human subjects were recruited or contacted. Because the analysis is restricted to already-public aggregate competition results without any personally identifiable information, it does not meet the standard threshold for human-subjects research under either domestic (e.g., MEXT / MHLW — the Ministry of Education, Culture, Sports, Science and Technology and the Ministry of Health, Labour and Welfare — ethical guidelines for research using publicly available secondary data) or international (e.g., 45 CFR §46.102 for non-human-subjects data) frameworks; accordingly, no institutional review board approval or waiver was sought or required.

Results

Descriptive statistics: three-era decomposition

The host-rank panel contains 150 prefecture-cup observations (75 non-cancelled editions × emperor's cup and empress's cup; the 1st, 2nd, 75th, and 76th editions are excluded from the analysis window for the reasons stated in Methods). Table 1 summarizes host top-1 / top-3 / top-8 rates by cup and by three eras (early 1948-1977, 30 editions/cup; golden 1978-2015, 38 editions/cup; shock 2016-2025, 7 editions/cup).

Two features stand out immediately. First, in the golden era the host prefecture won the emperor's cup in 37 of 38 editions (97.4%) and the empress's cup in 35 of 38 (92.1%); in the same era the host reached the top 3 in 37 of 38 editions on both cups (97.4%) and the top 8 in 37 of 38 (97.4%). Second, in the early era and in the shock era the top-1 host rate falls to comparable magnitudes on both cups (early 43.3% pooled across cups; shock 42.9%), while the top-3 and top-8 rates remain high (early top-3 = 75.0%, top-8 = 86.7%; shock top-3 = 100.0%, top-8 = 100.0%). The three-era time pattern is therefore non-monotonic on the top-1 metric — the host top-1 rate rises sharply from early to golden, then collapses back to an early-era-comparable level in the shock era — while remaining monotonically increasing (or roughly flat at the ceiling) on the top-3 and top-8 metrics. Figure 1 displays the individual host-rank observations across the full 1948-2025 window with era colour coding; Figure 2 displays the era × cup × threshold cell means with 95% Wilson two-sided confidence intervals.

Reality test: host top-k prevalence versus chance baseline

Table 2 reports the one-sample proportion test comparing observed host top-k rates against a chance-only null rate of k / 47 (top-1: 2.13%; top-3: 6.38%; top-8: 17.02%). Pooled across both cups (n = 150), the observed host top-1 rate is 0.693 (104 of 150) against a null of 0.021, an excess of 0.672; the exact-binomial one-sided upper-tail p is below any conventional threshold, so we report a shared safe cap of **$p < 10^{-50}$** for the pooled comparison (the true tail probability is far smaller; the safe cap is used throughout the paper for exponent-scale claims to avoid dependence on rounded numeric output). The corresponding top-3 pooled rate is 0.887 (133 of 150; excess 0.823) and the top-8 pooled rate is 0.933 (140 of 150; excess 0.763), both with the same safe cap.

Restricted by era, the pattern reappears with era-specific magnitudes: the early era pooled top-1 rate is 0.433 (26 of 60; excess 0.412), the golden era 0.947 (72 of 76; excess 0.926), and the shock era 0.429 (6 of 14; excess 0.407). The three era-restricted comparisons form a **separate secondary family** from the six primary confirmatory comparisons defined in Methods §Multiple comparisons; treating them as an additional Bonferroni family of three (α_single = 0.05 / 3 ≈ 0.017) or applying a joint Bonferroni across the union of both families (α_single = 0.05 / 9 ≈ 0.0056) makes no substantive difference here, because the smallest of the three era-restricted p-values (shock top-1 = 2.4×10^{-7}; see Table 2) sits far below either threshold. By cup, the observed host top-1 rates are 0.760 (emperor's cup, 57 of 75; Wilson 95% CI [0.652, 0.842]) and 0.627 (empress's cup, 47 of 75; Wilson 95% CI [0.514, 0.727]), each against a null of 0.021 — the exponent-scale claim ($p < 10^{-50}$) holds cup-by-cup and threshold-by-threshold. The observed excess above chance is present in every era × cup × threshold cell of Table 2 and is not the artifact of a single dominant era.

As a sensitivity check for the 1972 Okinawa-reversion period (when Japan comprised 46 prefectures rather than 47), we recomputed the early-era comparisons under a 46-state null (null rate = k / 46 rather than k / 47). Because null_rate = k / n_states is inversely proportional to n_states, the default 47-state null slightly overstates excess above chance for the pre-1972 window; under the 46-state null the observed excess differs from the 47-state baseline by less than 0.005 across every threshold-cup cell (top-1 tennou early: 0.5454 vs. 0.5449; top-3: 0.7695 vs. 0.7681; top-8: 0.6965 vs. 0.6928). The impact is therefore marginal and does not affect any qualitative reading of Table 2 (see Limitations and Supplement §S4 for the data appendix).

Permutation inference

To confirm that the observed excess is not a fixed-population artifact — for example, of a stable but heterogeneous prefecture-level top-k rate combined with a non-random host schedule — we conducted a permutation test that repeatedly reassigns the host label uniformly at random over the 47 prefectures within each edition and recomputes the top-k host rate under the shuffled labels (10,000 permutations; Table 3). The permutation-null top-k rates concentrate tightly around the theoretical chance rate (e.g., top-1 emperor's cup: null mean 0.021, null SD 0.017, null 95th percentile 0.053), and the observed host top-k rates exceed the 95th percentile of the permutation null by wide margins on every threshold × cup cell (top-1 emperor's cup: observed 0.760 vs. null 95th 0.053; top-3 empress's cup: observed 0.853 vs. null 95th 0.107). The empirical p-value from 10,000 permutations was zero on every cell; with the Phipson & Smyth (2010)[17] Monte Carlo lower-bound correction applied uniformly ((count + 1) / (n_perm + 1)), we report **p_perm = 1 / 10,001 = 9.999×10^{-5}** for all six threshold × cup cells. The permutation and one-sample proportion tests therefore agree on both direction and inferential strength: the observed host-prevalence excess is not explainable by chance under a within-edition random-host null.

Era heterogeneity: three-era χ^2 and pairwise Fisher

Table 4 reports the three-era χ^2 test with pairwise Fisher-exact follow-ups (both cups pooled to raise cell counts; Yates continuity correction inapplicable to 3 × 2 tables). On the top-1 metric, the pooled three-era comparison rejects the null of equal top-1 rate across eras at **$\chi^2 = 46.76$, df = 2, $p < 10^{-10}$**. The pairwise Fisher decomposition on the same pooled top-1 metric shows early-vs-golden $p < 10^{-10}$ (early 43.3% ≪ golden 94.7%) and golden-vs-shock $p < 10^{-5}$ (golden 94.7% ≫ shock 42.9%), while **early-vs-shock is**

essentially at unity (Fisher p = 1.00) — the shock era's top-1 host rate is statistically indistinguishable from the early era's rate. This is the formal test of the non-monotonic time pattern noted at the descriptive level: the shock era is not a simple return to a pre-golden baseline followed by continued decline but a return to the *statistically same* level. The top-3 and top-8 metrics show the same qualitative pattern — golden era ≫ early era and shock era ≈ early era — but at attenuated significance because the top-3 and top-8 rates are already near-ceiling in the early era.

Pooled ordered logit

Table 5 reports the pooled ordered-logit specification on host rank (censored at 9 = outside top 8) with era and cup dummies as fixed effects and cluster-robust standard errors at the host-prefecture level (46 unique host prefectures across the 75-edition sample; Tokushima is the only prefecture that has never served as primary host and appears in the panel only as a co-host row of the 8th and 48th editions). The pooled specification ($n = 150$) recovers a golden-era coefficient of **−3.33** ($SE = 0.86$, $p < 0.001$; negative coefficient corresponds to better numerical rank on the ordered scale) and a shock-era coefficient of **−0.54** ($SE = 0.54$, $p = 0.317$), both relative to the early-era reference. The cup coefficient (emperor's cup vs. empress's cup reference) is −0.89 ($SE = 0.27$, $p = 0.001$), consistent with the descriptive finding that host prefectures place better on the emperor's cup than on the empress's cup at every threshold. The cup-mode-specific ordered logits (emperor's cup only $n = 75$; empress's cup only $n = 75$) recover the same qualitative pattern, with golden-era coefficients of −3.37 (emperor's cup) and −3.45 (empress's cup) and shock-era coefficients of +0.08 (emperor's cup, $p = 0.89$) and −1.15 (empress's cup, $p = 0.045$). Only 7 of the 9 potential rank cells (1, 2, 3, 4, 5, 8, and 9) are actually observed in the pooled panel (rank 6 and rank 7 are unobserved across the full 150 rows), so the ordered-logit threshold parameters between rank 5 and rank 8 span three latent thresholds that the model does not separately identify; this is a numerical characteristic of the observed data-generating process rather than a fit failure.

Balmer et al. (2003)-style sport-type decomposition (2024-2025 subsidiary)

Because sport-level score data are only publicly available for the 2024 Saga and 2025 Shiga editions, the Balmer et al. (2003)-style objective/subjective decomposition is estimated as a subsidiary two-year cross-section rather than as an extension of the primary 75-completed-edition analysis. Table 6 reports the four internally consistent point estimates that this two-year design produces. The primary specification ($n = 6{,}991$ prefecture-sport-year-trophy cells, all 41 sports retained after handling edition-specific sport-catalog rotation between the two editions (the objective-side count changes from 17 to 16 sports across 2024→2025 because clay-target shooting was appended in 2024 and removed in 2025; the subjective- and semi-subjective-side counts are unchanged across the two editions); 47 prefecture clusters with 2 treated) recovers a host main effect of **+16.60 points per sport** ($SE = 1.25$, $p < 10^{-30}$) and a **host × subjective interaction of +16.68 points** ($SE = 7.72$, cluster-robust **$p = 0.031$**). Because the two-year window provides only 2 treated prefecture clusters (Saga in 2024 and Shiga in 2025) out of 47, the cluster-robust variance formula is known to be anti-conservative in this regime (Cameron & Miller, 2015[13]; MacKinnon & Webb, 2017[15]). We therefore accompany the cluster-robust p with a wild-cluster bootstrap (Rademacher weights, restricted null $\beta_{HS} = 0$, B = 999 iterations; Cameron, Gelbach & Miller, 2008[14]), which returns **bootstrap p = 0.155** and a pivotal 95% CI of [−0.42, +33.86]. Under the wild-cluster bootstrap, the interaction does not attain conventional significance, so we frame the +16.68 point estimate as directional (positive) evidence for the Balmer et al. (2003) decomposition rather than as a decisive rejection of the null.

Two internal robustness checks on the same 2024-2025 cross-section corroborate the direction of the primary interaction estimate. First, on a log-linear scale (dependent variable log(1 + score), inclusive n = 6,991), the interaction remains positive (**+0.293**, SE = 0.170, cluster-robust p = 0.084), directionally aligned with the primary linear specification. Second, a classification-variant sensitivity re-estimates the primary specification under a stricter delimitation of the subjective category. The “combat-excluded” variant, in which subjectivity is narrowed to pure artistic scoring (gymnastics and equestrian only; 9 combat sports treated as objective), yields **β_{HS} = +31.81** (SE = 8.20; cluster-robust $p < 10^{-3}$; wild-cluster bootstrap p = 0.174 on 2 treated clusters, close to and slightly larger than the primary specification’s bootstrap p = 0.155). All four Table 6 rows preserve $\beta_{HS} > 0$ across the plausible range spanned by inclusive, log-outcome, and combat-excluded specifications. The complementary classification variants (pure-judged; combat-to-semi) are documented in the analysis code and are consistent with the same directional finding, though their inferential status under the two-treated-cluster bootstrap correction is the same as the primary specification’s (bootstrap p in the 0.15-0.65 range).

Descriptively, the raw host-versus-nonhost per-sport score gap increases monotonically across the three Balmer et al. (2003) sport categories in exactly the direction the decomposition predicts (Table 7; Figure 3): objective sports show a gap of **+17.60 points** (host mean 43.63 vs. non-host mean 26.03, n_host = 65 cells across 17 sports), semi-subjective / team sports **+26.18 points** (47.03 vs. 20.85, n_host = 48 cells across 13 sports), and subjective sports **+37.91 points** (57.24 vs. 19.33, n_host = 38 cells across 11 sports). The subjective host boost of +37.9 is 2.15× the objective category’s +17.6, close to the doubling that the primary Table 6 point estimate implies for the average subjective sport relative to the average objective sport. Figure 3 displays the three category-specific host boosts with 95% confidence intervals from per-category `score ~ is_host` ordinary least squares (OLS) with prefecture-clustered SEs (47 clusters).

Cross-national context

Table 8 places the Kokutai host bonus in cross-national context. Among the four cases summarized, Japan is the clearest example in which a large host bonus (host top-1 rate 76.0% on the emperor’s cup pooled across 75 editions vs. a k / 47 chance baseline of 2.13%) emerges *without* any institutional score bonus in JSPO scoring rules — in contrast to Korea’s KSOC national comprehensive sports competition, which explicitly adds a 20% score bonus to the host city[8], and in contrast to the ~45% attenuation of the Olympic host coefficient once population and per-capita GDP are added as country-level controls (Csurilla & Fertő, 2023)[9]. The Chinese National Games host effect (Lan & Yu, 2012)[11] is characterized qualitatively rather than as an explicit institutional bonus, but the five descriptive attributes Lan & Yu list (high efficacy, gradual increase, delayed effect, exclusivity, double-edgedness) describe the same empirical phenomenon we observe at the Kokutai in more diffuse language.

Discussion

The Balmer et al. (2003) decomposition is directionally supported at the Kokutai

The central finding of this study is that the Kokutai host bonus appears to be directionally non-uniform across sport types: the marginal bonus in subjectively judged sports is directionally consistent with being roughly twice as large as the bonus in objectively measured sports. In the 2024-2025 stacked cross-section, the primary specification ($n = 6{,}991$ inclusive prefecture-sport-year-trophy cells, all 41 sports retained after handling edition-specific sport-catalog rotation between the two editions (the objective-side count changes from 17 to 16 sports across 2024→2025 because clay-target shooting was appended in 2024 and removed in 2025; the subjective- and semi-subjective-side counts are unchanged across the two editions); 47 prefecture clusters with 2 treated) yields a host main effect of +16.60 points per sport and a host × subjective interaction of +16.68 points (cluster-robust $p = 0.031$; wild-cluster bootstrap $p = 0.155$ under the few-treated-clusters correction on 2 treated prefectures out of 47), producing a model-implied total per-sport bonus of roughly 33 points in subjective sports against roughly 17 in objective ones — directionally consistent with a subjective-sport host bonus roughly 2× the objective one. The raw descriptive gap orders the three sport categories monotonically as +37.9 (subjective) > +26.2 (semi-subjective) > +17.6 (objective) — precisely the ordering that Balmer, Nevill & Williams (2003) documented at the summer Olympics. Given that the wild-cluster bootstrap does not reach conventional significance under the two-treated-cluster structure of this cross-section, we frame this as a *directionally* consistent first quantitative anchor for the Balmer et al. (2003) decomposition on Kokutai data — supported concordantly by (a) the raw descriptive monotonicity, (b) the log-outcome interaction direction, and (c) the linear interaction point estimate — rather than a decisive statistical confirmation. The Cameron-Gelbach-Miller (2008) pivotal wild-cluster bootstrap 95% confidence interval of [−0.42, +33.86] further makes the marginal-non-significance explicit at the interval level: the interval crosses zero by essentially the width of a rounding tolerance, consistent with the bootstrap p reported in Table 6 (0.155) but preserving the substantial positive-mass concentration of the point estimate. It nevertheless closes the specific gap left by Funahashi, Hibino, Ishiguro & Mano (2016), whose 2003-2011 panel documented a robust overall host bonus but did not disaggregate by sport type.

A three-variant classification sensitivity analysis further supports the invariance of the sport-type gradient to reasonable classification choices (Table 6). The combat-excluded variant — which restricts the subjective category to pure artistic scoring (体操 gymnastics + 馬術 equestrian, dropping the nine combat sports from the analysis frame entirely rather than reclassifying them as objective, and additionally dropping the 13 semi-subjective/team sports to isolate a pure objective-vs-subjective contrast (see `src/analysis_cross_section_2024_2025.py::run_bootstrap_by_variant` L553-554 for the exact filter chain and Table 6 row (iv) footnote for the sample-size accounting) — yields the largest cluster-robust point estimate on this data (β_HS = **+31.81**, SE = 8.20, cluster-robust **$p < 0.001$**; wild-cluster bootstrap $p = 0.174$, close to and slightly larger than the primary specification's bootstrap $p = 0.155$ under the few-treated-clusters correction applied consistently across all four variants), which is consistent with a pattern in which host bias appears strongest under pure artistic scoring rather than under combat-referee judgment (though this cross-variant comparison is a directional observation, not a formal test, since the combat-excluded variant's bootstrap $p = 0.174$ remains above conventional significance thresholds under the same few-treated-clusters correction). The complementary pure-judged and combat-to-semi variants (documented in the analysis code) yield β_HS = +18.99 and +26.40, respectively; the primary inclusive specification's β_HS = +16.68 together with the three classification variants (+18.99, +26.40, +31.81) preserve β_HS > 0 across the range +16.68 to +31.81, so the qualitative directional finding is invariant to how the subjective category is delimited within the plausible range spanned by Balmer et al. (2003)-consistent classification schemes.

The finding is consistent with, but does not require, a judging-bias mechanism. The Balmer et al. (2003) decomposition is an *implicit* test — an interaction of the observed type is what a judging-bias mechanism would produce, but a purely crowd-effect mechanism that is stronger in enclosed, spectator-dense subjective-sport venues (typical: judo, kendo, gymnastics halls) than in open-air objective-sport venues (typical: track and field, cycling road courses) would produce the same pattern. Without a judge-level dataset we cannot uniquely identify judging bias as the mechanism. What we can say is that a purely uniform material mechanism would be less likely to produce this sport-type gradient, although sport-specific preparation, venue familiarity, and transfer-athlete channels cannot be ruled out. The interaction we recover is more consistent with a crowd-and-judging pathway than with a purely uniform material one. Nomura (2022)[10] provides collateral evidence from Japanese J1 football that home advantage disappeared during the reduced-occupancy 2020 COVID-19 season and — through multiple-group structural equation modeling — that the crowd's influence on referees' decisions vanished under the same conditions, further supporting the crowd–referee pathway. This analogy from J1 to the Kokutai is inherently indirect: the two Kokutai editions that would have been the natural in-domain analogues of Nomura's reduced-occupancy 2020 J1 season — the 75th (2020 Kagoshima) and 76th (2021 Mie) editions — were cancelled for COVID-19 (see Methods and Limitations), so the crowd-referee mechanism cannot be tested at the Kokutai directly and the J1 evidence is offered as suggestive rather than confirmatory for Kokutai-specific inference.

Two host-loss clusters, not one

The popular framing of a "post-2016 Tokyo winning streak" is factually incorrect. Host prefectures won both 2018 Fukui (2,896 vs. Tokyo's 2,246) and 2019 Ibaraki (2,569 vs. Tokyo's 2,217) under exactly the same national scoring rules that produced the Tokyo wins in 2016 Iwate and 2017 Ehime. The 2020 and 2021 editions were cancelled for COVID-19, with the 2020 Kagoshima program re-scheduled as a special edition in 2023. Within the 2016-2025 shock era (7 non-cancelled editions × 2 cups = 14 host-cup cells), the host top-1 rate is 42.9% (6 of 14; both cups pooled), reflecting host losses concentrated in 2016 Iwate, 2017 Ehime, 2022 Tochigi, and 2024 Saga, interspersed with host wins in 2018 Fukui, 2019 Ibaraki, and 2025 Shiga rather than a uniform post-2016 host-loss regime. The 2002 Kōchi observation — the sole edition on which the primary host finished outside the top 8 (rank > 8) outside the early era, in the middle of the 1978-2015 golden era — sits apart from these clusters and is documented separately in Figure 1. The two-layer event-study specification we adopt (see Supplementary Materials §S1) preserves this within-shock-era heterogeneity as a design check on the three-era decomposition; a naive single-break DID at 2016 would produce a biased estimate of the structural change. This suggests a modeling recommendation for future Japanese work on the Kokutai host effect: 2016 should not be treated as a single structural break, and a framing that preserves the interspersed 2018/2019/2025 host wins within the broader 2016-2025 shock era is likely to be more informative.

East-Asian institutional context

The Kokutai host bonus emerges *in the absence of* an explicit institutional host bonus in JSPO scoring rules (Table 8 summarizes the cross-national institutional context). This stands in contrast to the Korean Sport & Olympic Committee's national comprehensive sports competition, which explicitly adds a 20% score bonus to the host city (10% from 2001, raised to 20% in 2010),[8] and to the qualitative characterization Lan and Yu (2012)[11] provide of the Chinese National Games host effect, whose five listed characteristics — high efficacy, gradual increase, delayed effect, exclusivity, and double-edgedness — describe roughly the same phenomenon we observe empirically at the Kokutai but in more diffuse qualitative language (this reference is retained at a *Low confidence* level: the source is a Chinese-local journal not indexed on Crossref, and only the abstract-level Chinese-language summary was accessible to us via a secondary-source portal; independent verification of the five-characteristics enumeration through the primary text was not possible with tools available to us, and the reference should be read in the "descriptive parallel" register rather than as a substantive citation). The present study is arguably an implicit decomposition of Lan & Yu's "exclusivity" and "double-edgedness" characteristics: exclusivity because the host bonus is concentrated in a subset of sports (subjectively judged), and double-edgedness because a large host bonus in a subjective sport increases both the host's expected medal count and the perceived legitimacy risk of that medal (see the Suetsugu 2024/2025 qualitative critique below).

Overlap and boundary with Suetsugu (2024, 2025)

The Suetsugu papers[4,5] provide a karate-specific qualitative critique of Kokutai outcomes, using the "host-victory-first mindset" framing (a phrase corresponding to the "開催地優勝至上主義" title of Suetsugu 2025) in a single-sport context. Our whole-sport panel-quantitative decomposition is complementary rather than overlapping: Suetsugu's papers focus on a single subjectively judged sport (karate) and characterize the outcome pattern in that sport in normative terms; our analysis provides the sport-level statistical evidence that the pattern Suetsugu describes in karate is indeed sport-type-specific (larger in subjectively judged sports than in objectively measured ones) rather than merely idiosyncratic to karate.

The two contributions can stand side by side: Suetsugu characterizes the qualitative concern within a single sport; we provide quantitative evidence consistent with the same interaction pattern across all sports.

We deliberately avoid the normative language Suetsugu uses. The observed interaction is consistent with judging bias, but it is also consistent with subject-sport-specific crowd effects, subject-sport-specific practice-environment effects, and subject-sport-specific transfer-athlete effects. The present study does not adjudicate among these mechanisms; it provides quantitative evidence consistent with their collective effect being larger in subjectively judged sports than in objectively measured ones, by a factor of roughly 2× under the primary specification.

The identified interaction is therefore a reduced-form estimate that lumps judge-bias, crowd-effect, practice-environment, and transfer-athlete channels; the absence of a JSPO judge-of-record roster (see Limitations, Second) prevents attribution of the observed effect to any single channel. A judging-bias mechanism, if present, would plausibly operate through host prefectures that supply larger judge cohorts for subjectively judged sports (gymnastics, kendo, karate, etc.) than for objectively measured sports, but we have no data with which to verify or quantify this asymmetry. The reduced-form estimate reported here should therefore not be read as identifying judge-bias per se, but rather as bounding the combined magnitude of the four candidate channels above.

Robustness across specifications

The reality-test finding — that the host prefecture reaches the top-1, top-3, and top-8 at rates far exceeding the k / 47 chance baseline — replicates in every era × cup × threshold cell of Table 2 (see Results), and the 46-versus-47-state sensitivity check confirms that the excess above chance differs from the 47-state baseline by less than 0.005 across every early-era threshold-cup cell. Unlike the ≈45% attenuation that Csurilla & Fertő (2023)[9] report for the country-level Olympic host coefficient when population and per-capita GDP are added as controls, the Kokutai host bonus emerges *without* any institutional score bonus or obvious socioeconomic driver at the primary-host level (Table 8), consistent with the Kokutai host-prefecture rotation being fixed by JSPO decades in advance rather than an economically endogenous selection.

Future work

A companion paper (in preparation) extends the categorical framework of the present study by introducing a continuous subjectivity gradient — scored on a Balmer et al. (2003)-consistent rubric across the full 40+ Kokutai sport catalogue with inter-rater reliability reported — and a sport-level exploratory screen for host-overperformance using newly collected competition-level top-N placement data. The companion paper is intended to complement (rather than supersede) the present categorical decomposition: the categorical contrast established here provides the qualitative direction, and the gradient/screen analyses would provide a finer-grained mechanistic decomposition together with case-study treatments of specific borderline sports (剣道 kendo, 柔道 judo, 相撲 sumo) that the current three-category framework treats uniformly.

Limitations

Eleven limitations of the present study should be noted.

First, edition-level sport-by-prefecture score data are only publicly available for the 2024 Saga and 2025 Shiga editions. All earlier editions (through 2023 Kagoshima special) publish only prefecture-level overall standings, not the sport-by-prefecture breakdown. This restricts the Balmer et al. (2003)-style objective/subjective interaction test to a two-year cross-section (primary n = 6,991 cells across the full sport catalogue) with only **2 treated prefecture clusters (Saga, Shiga) out of 47**. Under the few-treated-clusters diagnosis (Cameron & Miller 2015; MacKinnon & Webb 2017)[13,15], the cluster-robust variance formula is expected to be anti-conservative in this regime, and we explicitly report the wild-cluster bootstrap p = 0.155 alongside the cluster-robust p = 0.031 in the Results section; the interaction magnitude should therefore be viewed as a first empirical anchor rather than a definitive magnitude. Year fixed effects in the cross-section are effectively a single dummy given the two-year window. If JSPO releases historical sport-by-prefecture breakdowns in the future, extending the interaction to a longer panel would substantially increase the treated-cluster count and tighten both the estimate and its inferential support. No formal outlier-handling procedure (winsorization, trimming, or influence-diagnostic exclusion) is applied to the continuous per-sport score outcome in the primary Table 6 specification; the log-outcome robustness check (Table 6 row iii) mitigates functional-form sensitivity to right-skew but is not a substitute for point-mass outlier diagnostics, and residual sensitivity to individual host-sport-year cells with unusually large boosts (e.g., a single-year subject-sport medal cluster in one of the two treated prefectures) is not separately quantified in this study.

Second, we could not obtain the JSPO judge-of-record roster (name + prefecture of affiliation) for any Kokutai edition. Four independent search paths — JSPO regulations catalog `tabid188`, seven individual-edition sampling, kendo-specific archives, and general web search — all returned negative. The Balmer et al. (2003)-style decomposition is therefore the only implicit test of judging bias we can offer; a direct judge-level Zitzewitz (2006)-style analysis[12] is not available for Kokutai data at present.

Third, the subjectively judged sports for which we could confirm high-fidelity edition-level score data (independent of the JSPO overall-standings pipeline) are essentially limited to kendo, which published a per-prefecture score table for the 2016 Iwate edition (71st) but discontinued the same table by 2022 Tochigi (77th). Judo, gymnastics, and other subjectively judged sports were not covered in this way. The 2024-2025 cross-section provides sport-level scores, but the pre-2024 subjective-sport history is patchy.

Fourth, the Suetsugu (2024, 2025) full texts are served through the Komazawa University institutional repository as WEKO3 octet-stream downloads that could not be reliably extracted; the citations are at bibliographic-plus-CiNii-link level only. This is a citation-quality limitation, not a data limitation, and does not affect the present study's numerical results.

Fifth, the Cabinet Office Economic and Social Research Institute (ESRI) 2008SNA prefectural GDP series that anchors the reference socioeconomic dataset for panel-based specifications is available only for fiscal 2011-2022 (令和4年度版) at the time of manuscript preparation. When ESRI releases the complete 47-prefecture 令和5年度版 (extending through fiscal 2023; a partial release covering 31 prefectures and 2 designated cities is currently available), a socioeconomic-controls-augmented panel can be extended by one year with essentially the same code path; when the 令和6年度版 releases (extending through 2024), such a panel can be extended to include the 2024 Saga edition (currently reported only in the 2024-2025 cross-section). Extending covariate coverage to 2024 would allow the 2024 Saga host-loss event to enter any future socioeconomic-controls-augmented analysis directly, tightening the identifying variation on the post-2016 cluster. All sample sizes reported in this manuscript (n = 150 host-rank main panel; n = 6,991 primary cross-section) are availability-driven — they reflect the maximum data currently released by JSPO and ESRI for each specification rather than a *ex ante* power-calculation design — so power/precision justifications are inherently tied to the JSPO/ESRI release schedule and would be strengthened by future releases of pre-2024 sport-by-prefecture breakdowns and post-2022 ESRI SNA series.

Sixth, we did not implement a formal proportional-odds (PO) diagnostic (e.g., a Brant-style partial-proportional-odds test) on the v3 pooled ordered logit specification (`rank_ordinal ~ C(era) + C(cup)`, n = 150 pooled across cups; Table 5). A preliminary Brant-style diagnostic run on a deprecated 2012-2022 legacy 47-prefecture-year panel (n = 423; archived in the git history at commit `fa07fd2` and earlier of the same GitHub repository — modules `src/analysis_main.py` and `src/analysis_confounders.py`, invoked via `run_all_models.py` in that commit — for reproducibility of this preliminary diagnostic, but the legacy panel and its implementation are not part of the present study's main analyses) showed convergence only at the top-1 threshold ($Y \leq 1$) and numerical singularity at all higher thresholds ($Y \leq 2$ through $Y \leq 8$), because every host prefecture-year in that legacy panel finishes in the top 8 (host_top8_rate = 100% on the 9 host rows), leaving no within-cell variance for those threshold-specific binary logits. This preliminary diagnostic cannot be directly ported to the v3 host-rank specification — which pools across 75 non-cancelled editions × 2 cups rather than across 47 prefectures within 9 years — but the same near-complete separation regime applies qualitatively: the host effect is so concentrated at the top rank that there is essentially no room for β_j to vary from threshold to threshold. A formal v3-specific proportional-odds diagnostic is left as future work. The substantive conclusions in this paper rest on the one-sample proportion tests (which do not depend on the PO assumption at all), the permutation tests, the era χ^2 comparisons, and the pooled ordered logit (whose PO assumption we cannot presently certify but which produces coefficients consistent with the descriptive top-1 pattern reported throughout the Results section), and are therefore not overturned by the untestable status of the PO assumption in this particular ordered-logit specification.

Seventh, the β_HS (host × subjective interaction) coefficient in the 2024-2025 cross-section rests on an asymmetric identification structure that is worth making fully explicit alongside the few-treated-clusters caveat already noted in the First limitation above. In practical terms, the interaction is identified off the difference between Saga and Shiga in their subjective-vs-objective host-cell contrasts. Under the primary (inclusive) specification, the interaction β_HS = +16.68 is identified off the roughly 38 subjective host cells (Saga 2024 subjective sports + Shiga 2025 subjective sports) contrasted against the roughly 48 semi-subjective host cells and 65 objective host cells within the same two prefectures. This is not a generic "2 of 47" cross-section design; it is a specific 2-prefecture design in which the interaction's variance is a function of exactly how those two prefectures happened to differ in subjective-sport versus objective-sport performance during their respective host years. If Saga and Shiga had turned out to be substantively similar in their subjective-sport performance (a plausible counterfactual given both are mid-sized prefectures without an obviously dominant elite-athlete pipeline in judo/kendo/gymnastics), the interaction would be estimated off a small residual signal and the true SE would be correspondingly inflated. This asymmetric identification is one of the mechanical reasons the wild-cluster bootstrap p (0.155) diverges substantially from the naive cluster-robust p (0.031) — the bootstrap re-samples over prefecture clusters and honestly reflects the fragility of a two-cluster identification base, whereas the cluster-robust SE leans on asymptotic approximations known to be anti-conservative in this regime (Cameron & Miller 2015; MacKinnon & Webb 2017)[13,15]. Extending the interaction test to a longer panel with more host prefectures would replace the current pairwise Saga-vs-Shiga structure with a broader within-prefecture identifying variance and correspondingly relax both the anti-conservative-SE concern and the two-prefecture asymmetric-identification concern.

Eighth, the wild-cluster bootstrap reported in the primary specification (Table 6) and across the three additional classification variants (Table 6) uses Rademacher weights (±1 with equal probability). With only two treated prefecture clusters (G_1 = 2), the discrete Rademacher weight distribution places non-trivial mass at just four possible sign patterns for the treated clusters (++, +-, -+, –), so the bootstrap p distribution may exhibit lumpiness at the tail-mass boundary. As a robustness check, we re-estimated the primary specification bootstrap using Mammen (1993) two-point continuous weights[16] (reported in Supplementary Materials §S2); the Mammen-weighted p-value 0.181 is qualitatively identical to the Rademacher-weighted value 0.155, confirming that the reported bootstrap inference does not depend on the choice of weight distribution within the wild bootstrap family. The Rademacher-weight discreteness is therefore not driving the reported non-significance under the few-treated-clusters correction.

Ninth, the one-sample proportion tests reported in the Results section use the default null rate k/47 based on the current 47-prefecture composition. During the early era (1948-1977, editions 3-32), 24 of the 30 non-cancelled editions preceded the 1972 reversion of Okinawa, when only 46 prefectures participated. Under a 46-state null applied uniformly to the early-era 30 editions as a conservative worst-case bound (pre-Okinawa reversion), the excess above chance differs from the 47-state baseline by less than 0.005 across all threshold-cup cells (top-1 tennou: 0.5454 vs. 0.5449; top-3 tennou: 0.7695 vs. 0.7681; top-8 tennou: 0.6965 vs. 0.6928); the default 47-state null therefore mildly overstates excess but the impact is marginal (see Supplementary Materials §S4).

Tenth, the JSPO official retrospective volume `01_kokutai.pdf` (692 pages, covering editions 1-65) contains competition-level top-6 placements per sport per edition but was not incorporated into the present sport-level analysis. Five obstacles blocked structured parsing: (i) a three-column layout that fragments same-competition results across non-adjacent lines, (ii) top-6 as the volume-wide standard, which is inconsistent with the top-8 threshold family used in the present study's main panel, (iii) frequent tie placements (multiple ⑤ entries per event), (iv) individual events reporting the competitor's affiliation as a club, school, or corporate team name (e.g., 門鉄, 菊花女専, 立命大, 稲泳会) rather than a prefectural affiliation, precluding prefecture-level aggregation for a non-trivial share of records, and (v) format drift across the 65-edition window that would require per-era parsing rules. Extending the sport-level analysis to editions 1-65 through structured extraction of this volume is left as future work.

Eleventh, the 80th edition (2026 Aomori) was not yet held as of manuscript preparation. The Kokutai (renamed the National Sports Games as of the 78th edition) is typically held in the autumn (September-October), so the 80th edition's rank data may or may not be publicly available at the time of SSRN submission. If the 47-prefecture overall standings for the 80th edition are published before submission, the main-panel analysis (host-rank pipeline) can be extended by one edition with essentially the same code path; if the data are not yet available, the present analysis remains fixed at the 75 completed editions in the 3-79 span (77-edition span with the 2020 and 2021 editions cancelled for COVID-19) and this 80th-edition limitation is retained.

Conclusion

Japan's National Sports Festival shows a host bonus that appears directionally non-uniform across sport types. The results are directionally consistent with a larger host bonus in subjectively judged sports than in objectively measured sports under the primary (inclusive) specification retaining all 41 sports ($n = 6{,}991$: host × subjective interaction = +16.68 points on a host main effect of +16.60 points on the combined objective + semi-subjective/team reference category, i.e., a model-implied per-sport subjective bonus that, added to the reference-category main effect, is roughly 2× the reference bonus; cluster-robust $p = 0.031$; wild-cluster bootstrap $p = 0.155$ under the few-treated-clusters correction on 2 treated prefectures out of 47; Mammen-weighted bootstrap $p = 0.181$; Cameron-Gelbach-Miller pivotal 95% CI [−0.42, +33.86]), directionally matching the Balmer, Nevill & Williams (2003) decomposition established at the summer Olympics. The overall Kokutai host bonus emerges in the absence of any institutional score bonus (unlike the Korean KSOC's 20% host multiplier), and stands in qualitative contrast to Csurilla & Fertő's (2023) finding that the Olympic host coefficient attenuates by roughly 45% once population and per-capita GDP controls are introduced; a formal prefecture-level population + GDP controls analysis for the Kokutai host bonus is not undertaken here because the ESRI 2008SNA prefectural GDP series covers only 2011-2022 at the time of manuscript preparation (see Limitations §Fifth) and is left as future work. The structural change in host championship rate around 2016 is best characterized as two clusters of host-loss editions — 2016 Iwate, 2017 Ehime, 2022 Tochigi, and 2024 Saga — interspersed with host wins in 2018 Fukui, 2019 Ibaraki, and 2025 Shiga, rather than a single break at 2016. The finding closes the specific quantitative gap left by Funahashi et al. (2016) — whose 2003-2011 panel documented an overall host bonus but did not disaggregate by sport type; a Funahashi-specification replication on the 2003-2011 panel is retained in Supplementary Materials §S3 for pipeline validation — and provides Japanese panel-econometric evidence complementary to the qualitative critique of Suetsugu (2024, 2025). Whether the observed interaction reflects judging bias, subject-sport-specific crowd effects, or subject-sport-specific practice-environment effects cannot be uniquely identified from the present design; the interaction is directionally supported by the descriptive monotonicity, the log-outcome robustness check, and a three-variant classification sensitivity in which the primary specification ($\beta_{HS} = +16.68$) together with the three classification variants (+18.99 pure-judged; +26.40 combat-to-semi; +31.81 combat-excluded) preserve $\beta_{HS} > 0$ across the range +16.68 to +31.81 (Table 6). The combat-excluded variant's cluster-robust $p < 10^{-3}$ shares the same two-treated-cluster limitation as the primary specification — its wild-cluster bootstrap p (0.174) is close to and slightly larger than the primary's (0.155), so its inferential status under the few-treated-clusters correction is the same as the primary — and the whole sport-type interaction result should be read in the "first empirical anchor" register that longer panels could sharpen into a decisive test.

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Acknowledgments

Claude Opus 4.7 (Anthropic) was used to assist with code drafting, debugging, literature-search support, and manuscript editing. All data extraction, statistical outputs, and numerical results were independently checked by the author by re-executing the analysis code (Python 3.14; unit tests n = 326, all passing) and cross-referencing outputs against the tabulated results. The author is solely responsible for the accuracy of the numerical results, the choice of specifications, the interpretation of coefficients, and the conclusions.

All references were verified against CrossRef, J-STAGE, PubMed, and PMC records where applicable; bibliographic verification records archived with the analysis code document the verification process, including two corrections to entries that appeared in earlier drafts (Funahashi et al. 2016 co-author list and journal; Csurilla & Fertő 2023 author count).

Author Contributions (Contributor Roles Taxonomy, CRediT)

Mizuki Shirai: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Software, Writing — Original Draft, Writing — Review & Editing, Visualization, Project Administration.

Conflict of Interest

The author declares no conflicts of interest per International Committee of Medical Journal Editors (ICMJE) guidelines.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data Availability

All input data used in this study are publicly available. Prefecture rankings are from the Nagano Prefecture Sports Association (https://www.nagano-sports.or.jp/kokutai/record/high_rank.html); individual-edition PDFs and 2024-2025 Excel breakdowns are from the JSPO official archive (https://www.japan-sports.or.jp/kokutai/tabid183.html); socioeconomic covariates are from the Cabinet Office ESRI prefectural accounts (https://www.esri.cao.go.jp/jp/sna/data/data_list/kenmin/files/contents/main_2022.html).

Analysis code (Python 3.14, pandas 3.0.5, statsmodels 0.14.6, matplotlib 3.11.1, weasyprint 69.0) — including the data-assembly scripts, all v3 regression modules (v3 pooled ordered logit on the 75-completed-edition host-rank panel (span 3-79), Funahashi replication OLS, 2024-2025 cross-section), the wild-cluster bootstrap implementation, the plotting pipeline, and the PDF build pipeline — is publicly available at https://github.com/rehabilitation-collaboration/kokutai-home-advantage (MIT License) and will additionally be permanently archived on Zenodo (with a DOI) concurrent with SSRN posting. The deprecated 2012-2022 legacy 47-prefecture-year Csurilla-style staged-confounder implementation, the pooled ordered/binary logit implementation on that legacy panel, and the Brant-style partial-proportional-odds diagnostic implementation used for the preliminary diagnostic reported in Limitations §Sixth and for the pipeline validation reported in Supplementary Materials §S3 Extension through 2022 are archived in the git history at commit `fa07fd2` and earlier of the same GitHub repository (modules `src/`

`analysis_main.py` and `src/analysis_confounders.py`, invoked via `run_all_models.py` in that commit). This commitment is made explicit to avoid an asymmetric-disclosure gap in which AI-usage transparency (see Acknowledgments — LLM-assisted development is fully disclosed at the point of first posting) would run ahead of code transparency and thereby impair independent reviewer verification of the LLM-assisted numerical results reported here.

Tables

Table 1. Host top-k rates by era × cup × threshold (v3 primary; n = 150 = 75 editions × 2 cups)

Era	Cup	n editions	n top-1	n top-3	n top-8	Rate top-1	Rate top-3	Rate top-8
early (1948-1977)	emperor's	30	17	25	26	0.567	0.833	0.867
early (1948-1977)	empress's	30	9	20	26	0.300	0.667	0.867
golden (1978-2015)	emperor's	38	37	37	37	0.974	0.974	0.974
golden (1978-2015)	empress's	38	35	37	37	0.921	0.974	0.974
shock (2016-2025)	emperor's	7	3	7	7	0.429	1.000	1.000
shock (2016-2025)	empress's	7	3	7	7	0.429	1.000	1.000
all (1948-2025)	emperor's	75	57	69	70	**0.760**	**0.920**	**0.933**
all (1948-2025)	empress's	75	47	64	70	**0.627**	**0.853**	**0.933**

Panel: 150 prefecture-cup observations = 75 non-cancelled editions × emperor's cup and empress's cup. Editions excluded: 1st (1946) and 2nd (1947) — no official 47-prefecture ranking available; 75th (2020 Kagoshima) and 76th (2021 Mie) — cancelled for COVID-19 (see Methods). Three multi-prefecture co-hosted editions (7th, 8th, 48th) are normalized to their primary host (see Methods and Data Availability). The three-era decomposition uses ERA_BOUNDARIES = {early: 1948-1977, golden: 1978-2015, shock: 2016-2025}. The non-monotonic pattern on the top-1 metric (early 0.433 pooled → golden 0.947 pooled → shock 0.429 pooled) is the descriptive fingerprint of the golden-era spike sandwiched between the early and shock eras; the pattern is formally tested in Table 4.

Table 2. One-sample proportion test: host top-k rate vs. chance baseline k/47 (both cups pooled; six primary confirmatory rows)

Era	Threshold	n	n_success	Observed rate	Null rate (k/47)	Excess	Wilson 95% CI (two-sided)	Exact-binomial upper-tail p
all (1948-2025)	top-1	150	104	0.693	0.021	0.672	[0.615, 0.762]	$< 10^{-50}$ (safe cap; exact ≈ 4.8×10^{-136})
all (1948-2025)	top-3	150	133	0.887	0.064	0.823	[0.826, 0.928]	$< 10^{-50}$ (safe cap; exact ≈ 4.2×10^{-138})
all (1948-2025)	top-8	150	140	0.933	0.170	0.763	[0.882, 0.963]	$< 10^{-50}$ (safe cap; exact ≈ 4.0×10^{-94})
early (1948-1977)	top-1	60	26	0.433	0.021	0.412	[0.316, 0.559]	1.2×10^{-27}
golden (1978-2015)	top-1	76	72	0.947	0.021	0.926	[0.872, 0.979]	$< 10^{-50}$ (safe cap; exact ≈ 4.8×10^{-115})
shock (2016-2025)	top-1	14	6	0.429	0.021	0.407	[0.214, 0.674]	2.4×10^{-7}

Six primary confirmatory rows; Bonferroni-corrected α_single = 0.05 / 6 ≈ 0.0083 is exceeded by every row. Exact-binomial one-sided upper-tail p-values are computed from scipy.stats.binomtest with alternative='greater'; the safe-cap notation `p < 10⁻⁵⁰` *is used for rows whose true tail probability sits below* 10^{-50} *so that the exponent is not tied to rounded numeric output (M4-A safe-cap policy). Rows with n = 60, n = 76, and n = 14 report the true exact tail in scientific notation because their tail probabilities exceed* 10^{-50}*. Additional cup-restricted, threshold-restricted, and era-restricted cells (36 rows in total) are documented in* `results/analysis_main_v3.txt` *[2] and preserve the same qualitative conclusion at every cell. A 46-state sensitivity check (early era, pre-Okinawa reversion) is discussed in the accompanying text and Limitations; the excess above chance differs from the 47-state baseline by less than 0.005 across every threshold-cup cell (data appendix deferred to Supplement §S4).*

Table 3. Permutation test: host_pref reassigned uniformly at random within each edition (n_perm = 10,000; Phipson & Smyth 2010 Monte Carlo lower-bound correction)

Threshold	Cup	Observed rate	Null mean	Null SD	Null 5th percentile	Null 95th percentile	p_perm
top-1	emperor's	0.760	0.021	0.017	0.000	0.053	1 / 10,001 ≈ 9.999 × 10^{-5}
top-3	emperor's	0.920	0.064	0.028	0.027	0.107	1 / 10,001 ≈ 9.999 × 10^{-5}
top-8	emperor's	0.933	0.169	0.043	0.107	0.240	1 / 10,001 ≈ 9.999 × 10^{-5}
top-1	empress's	0.627	0.021	0.017	0.000	0.053	1 / 10,001 ≈ 9.999 × 10^{-5}
top-3	empress's	0.853	0.064	0.028	0.027	0.107	1 / 10,001 ≈ 9.999 × 10^{-5}
top-8	empress's	0.933	0.166	0.043	0.093	0.240	1 / 10,001 ≈ 9.999 × 10^{-5}

For each threshold × cup cell, 10,000 permutations reassign the host label uniformly at random over the 47 prefectures within each edition and recompute the observed top-k host rate under the shuffled labels. The empirical count of permutations reaching or exceeding the observed rate was zero across all six cells; the Phipson & Smyth (2010) Monte Carlo lower-bound correction `(count + 1) / (n_perm + 1)` *yields a common p-value of 9.999 × 10^{-5}. The permutation null concentrates tightly around the theoretical chance rate (null mean ≈ k/47), confirming that the shuffle preserves the fixed 47-prefecture within-edition structure without inducing extraneous variance.*

Table 4. Three-era χ² and pairwise Fisher exact (host top-k rate by era; Yates continuity correction inapplicable to 3 × 2 tables)

Cup	Threshold	Early rate	Golden rate	Shock rate	χ^2 (df = 2)	χ^2 p	Fisher early_vs_golden	Fisher golden_vs_shock	Fisher early_vs_shock
emperor's	top-1	0.567	0.974	0.429	19.88	4.8×10^{-5}	$< 10^{-4}$	1.1×10^{-3}	0.680
emperor's	top-3	0.833	0.974	1.000	5.16	0.076	0.080	1.000	0.560
emperor's	top-8	0.867	0.974	1.000	3.64	0.162	0.162	1.000	0.570
empress's	top-1	0.300	0.921	0.429	28.93	5.2×10^{-7}	$< 10^{-7}$	6.8×10^{-3}	0.659
empress's	top-3	0.667	0.974	1.000	13.95	9.4×10^{-4}	8.4×10^{-4}	1.000	0.155
empress's	top-8	0.867	0.974	1.000	3.64	0.162	0.162	1.000	0.570
both pooled	**top-1**	**0.433**	**0.947**	**0.429**	**46.76**	$\mathbf{7.1 \times 10^{-11}}$	$\mathbf{< 10^{-10}}$	$\mathbf{< 10^{-5}}$	**1.000**
both pooled	top-3	0.750	0.974	1.000	18.67	8.8×10^{-5}	$< 10^{-4}$	1.000	0.059
both pooled	top-8	0.867	0.974	1.000	7.27	0.026	0.022	1.000	0.339

The pooled top-1 row (highlighted) is the primary formal test of the non-monotonic time pattern. The three-era χ^2 comfortably rejects equal top-1 rates across eras ($\chi^2 = 46.76$, $p < 10^{-10}$), and the pairwise Fisher decomposition places both the golden-vs-early and the golden-vs-shock differences well below any conventional threshold while the early-vs-shock difference sits at Fisher $p = 1.000$ — an exact statistical identity in the two-by-two collapse of a 30-vs-14 sample with 26 and 6 successes. The top-3 and top-8 rows attenuate because the empress's-cup and emperor's-cup rates approach 100% in the golden and shock eras; the qualitative "golden ≫ early ≈ shock" pattern remains on all thresholds. Fisher-exact p-values reported to 4 significant figures when $> 10^{-3}$ and as safe-cap upper bounds $< 10^{-n}$ when the true tail exceeds numerical resolution.

Table 5. Pooled ordered logit (rank_ordinal ~ C(era) + C(cup); cluster-robust SE at host_pref_code, 46 unique host prefectures)

Cup mode	Variable	Coefficient	SE	z	p	95% CI (Wald)	n_obs
pooled (both cups)	era_golden (vs. early ref)	**−3.332**	0.862	−3.86	**1.1×10^{-4}**	[−5.02, −1.64]	150
pooled	era_shock (vs. early ref)	−0.540	0.540	−1.00	0.317	[−1.60, +0.52]	150
pooled	cup_tennou (vs. kougou ref)	−0.892	0.270	−3.30	**9.6×10^{-4}**	[−1.42, −0.36]	150
emperor’s cup only	era_golden	−3.368	1.201	−2.81	**5.0×10^{-3}**	[−5.72, −1.01]	75
emperor’s cup only	era_shock	+0.081	0.582	+0.14	0.890	[−1.06, +1.22]	75
empress’s cup only	era_golden	−3.449	0.881	−3.92	**8.8×10^{-5}**	[−5.18, −1.72]	75
empress’s cup only	era_shock	−1.153	0.574	−2.01	**0.045**	[−2.28, −0.03]	75

Reference categories: era = early (1948-1977) and cup = kougou (empress’s cup). Negative era coefficient corresponds to better (numerically smaller) host rank on the ordered scale. 95% Wald confidence intervals are computed as coefficient ± 1.96 × cluster-robust SE. Cluster-robust standard errors are computed at `host_pref_code` *(46 unique clusters; Tokushima is the only prefecture that has never served as primary host and appears only as a co-host row of the 8th and 48th editions). Threshold parameters (cut-points between adjacent observed ranks) are omitted for brevity; the observed ranks span only 7 of 9 latent categories (rank 6 and rank 7 are unobserved across all 150 rows), so the threshold spans between observed rank 5 and observed rank 8 encapsulate three latent thresholds that the ordered logit does not separately identify. Full model output (including thresholds) is documented in* `results/analysis_main_v3.txt` *[5]; the log-likelihoods are −126.15 (pooled), −48.19 (emperor’s cup), and −74.29 (empress’s cup).*

Table 6. 2024-2025 cross-section: Balmer et al. (2003)-style sport-type decomposition (2 treated prefectures out of 47; four internally consistent point estimates)

Row	Specification	Sample	β_H	β_HS	SE (β_HS)	Cluster-robust p (β_HS)	Bootstrap p (β_HS)	95% CI (β_HS)	R^2
(i)	**Primary: inclusive (all 41 sports)**, cluster-robust view	n = 6,991	+16.60	**+16.68**	7.72	**0.031**	—	[+1.55, +31.81] cluster	0.244
(ii)	Primary: inclusive, wild-cluster bootstrap view (Rademacher, B = 999, restricted null)	n = 6,991	+16.60	**+16.68**	7.72	0.031	**0.155**	[−0.42, +33.86] bootstrap pivotal	0.244
(iii)	Log-outcome robustness: log(1 + score), inclusive, cluster-robust	n = 6,991	+0.467	+0.293	0.170	0.084 (marginal)	—	[−0.040, +0.626] cluster	0.279
(iv)	Combat-excluded variant (obj vs. subj = 体操 + 馬術 only; 9 combat sports excluded and 13 semi-subjective/ team sports excluded to isolate a pure objective-vs-subjective contrast; classification-variant sensitivity)	n = 3,359	+10.71	**+31.81**	8.20	**1.0×10^{-4}**	**0.174**	not computed (variant runs through the bootstrap-only path in `run_bootstrap_by_variant`, which does not emit a cluster-robust CI; the pivotal bootstrap CI is available on request from the same seed but was not tabulated)	—

Rows (i) and (ii) are the same primary specification presented under two inferential views (cluster-robust and wild-cluster bootstrap) rather than two distinct specifications; the point estimates and cluster-robust p are identical. The four rows collectively demonstrate that (i)-(ii) the primary interaction is directionally positive under both inference regimes but does not attain conventional significance under the appropriate few-treated-clusters correction; (iii) the direction is preserved on a log-linear scale; (iv) a stricter delimitation of the subjective category (pure artistic scoring only, with combat sports and semi-subjective/ team sports both excluded from the analysis frame rather than reclassified) recovers the largest cluster-robust point estimate on this data, with the bootstrap p (0.174) matching the primary variant's within 0.001 under the two-treated-cluster correction applied consistently. The primary specification (rows i-ii) uses the inclusive n = 6,991 sample retaining all 41 sports (33 sports common to both 2024 Saga and 2025 Shiga plus 8 sports appearing in only one edition, of which clay-target shooting was appended in 2024 and removed in 2025); a semi-excluded primary variant (13 semi-subjective/team sports removed to isolate a pure obj-vs-subj contrast) yields β_H = +11.18, β_HS = +20.27, cluster-robust p = 0.027, wild-cluster bootstrap p = 0.175, and is documented in the analysis code (`src/analysis_cross_section_2024_2025.py`*). Row (iv)'s combat-excluded variant is implemented via* `run_bootstrap_by_variant` *with* `category_variant="no_combat"` *(drops the 9 combat sports from the analysis frame; L120-122 of the source module) followed by an additional filter* `df[df["is_semi"] == 0]` *(drops the 13 semi-subjective/team sports; L554 of the source module), leaving 19 sports (17 objective + 2 pure-artistic subjective = 体操 + 馬術) and n = 3,359 obs. Prefecture, sport, year, and trophy fixed effects included in all specifications; 47 prefecture clusters (Saga in 2024 and Shiga in 2025 are the two treated clusters), Rademacher weights on prefecture clusters for the bootstrap, seed 20260728 shared across all bootstrap variants. R^2 is the OLS coefficient of determination on the full 6,991-cell (or 3,359-cell) frame with all fixed effects and interactions; row (iv)'s R^2 is reported through the bootstrap runner rather than the OLS runner and therefore is omitted from this summary table (available in the analysis output).*

Table 7. Descriptive host-boost by Balmer et al. (2003) sport category (2024-2025 stacked cross-section; per-sport score)

Category	Mean non-host	Mean host	Cells (non-host)	Cells (host)	Host boost (points)
Objective (17 sports)	26.03	43.63	2,920	65	**+17.60**
Semi-subjective / team (13 sports)	20.85	47.03	2,199	48	**+26.18**
Subjective (11 sports)	19.33	57.24	1,721	38	**+37.91**

Monotonic increase in raw host boost across the three Balmer et al. (2003) sport categories, in exactly the direction the decomposition predicts. The subjective category's +37.91 is 2.15× the objective category's +17.60, close to the doubling ratio implied by the Table 6 primary specification (see main text). Figure 3 displays these three category-level host boosts with 95% confidence intervals from per-category `score ~ is_host` *OLS with prefecture-clustered SEs (47 clusters).*

Table 8. Cross-national context: host-scoring institutional arrangements

System	Host score bonus	Empirical host effect	Reference
Japan Kokutai (present study)	None (no institutional bonus in JSPO scoring)	Emperor's-cup host top-1 rate 76.0% (57 of 75) vs. chance 2.13%; +16.68 subjective interaction on a +16.60 objective base per sport (Table 6 primary; 2024-2025 subsidiary)	Present study
Korea national comprehensive sports	20% score bonus (10% from 2001, raised to 20% in 2010)	Institutionally amplified	Kim (2025), Munhwa Ilbo[8]
China National Games	(institutional bonus not documented in accessible sources)	Qualitatively described (five characteristics: high efficacy, gradual increase, delayed effect, exclusivity, double-edgedness)	Lan & Yu (2012)[11]
Summer Olympics	None	Attenuates ~45% under population + GDP + country FE	Csurilla & Fertő (2023)[9]

Among the four cases summarized here, Japan is the clearest example in which a large host effect emerges without any institutional score bonus and without an obvious socioeconomic driver at the primary-host level.

Supplementary Materials

§S1. Event-Study Design and Results

An initial single-break difference-in-differences (DID) centered at 2016 was rejected because the raw data show host wins in both 2018 (Fukui) and 2019 (Ibaraki), i.e., a return to the pre-2016 pattern within the "post" period — this violates the parallel-trend assumption of a single-break design. We instead specified a two-layer event-study following an exploratory diagnostic that separates the pre-2005 *furusato*-athlete-rule regime from the post-2016 regime: **Layer 1** (pre-2005 regime) contains the single 2002 Kochi host-loss event as an isolated pre-*furusato*-athlete-rule (introduced 2005) shock, and **Layer 2** (post-2016 regime) stacks the five host-loss events of 2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima (delayed special edition), and 2024 Saga. The event-study is fitted as a linear-probability model (top-1 outcome) with prefecture-clustered SEs; formal parallel-trend Wald tests are reported.

The parallel-trend Wald tests were formally non-rejecting in both layers (Wald $p = 1.0$), but the identifying variation is thin: the treated group (host prefectures) exhibits non-zero top-1 status essentially only in the host year itself, and the pre- and post-shock cells for the treated group are mechanically zero. As a result, the event-study point estimates are near-zero by construction and should be read as a design check rather than as an independent identifying analysis of the shock timing. Supplementary Figure S1 displays the τ-coefficient trails for both layers alongside their 95% confidence bands. This limitation is a fundamental consequence of the top-1 outcome variable being effectively a host-year indicator; a continuous-score outcome would in principle allow a stronger event-study, but the 2003-2011 replication window is where continuous-score data are available, and that window contains no post-2016 shocks.

Supplementary Figure S1. Two-layer event-study τ-coefficient trails on top-1 (championship) status. Layer 1: 2002 Kochi single host-loss event (pre-2005 *furusato*-athlete-rule regime). Layer 2: post-2016 stacked five host-loss events (2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima delayed-special edition, and 2024 Saga). *Estimation.* Linear-probability model on top-1 outcome with prefecture-clustered SEs (47 clusters); $\tau = 0$ aligns to the host-loss year of each treated prefecture; treated pool spans $\tau \in [-3, +3]$ within each layer where the corresponding calendar year lies in the layer's regime. *Reading the τ coefficients.* All τ point estimates lie within ±0.10 of zero with wide 95% confidence bands, which reflects the mechanical consequence — not evidence for or against the shock hypothesis — that the top-1 outcome variable is essentially a host-year indicator: a treated prefecture registers a top-1 event only in its own host year, so its pre-$\tau = 0$ and post-$\tau = 0$ cells are structurally zero, leaving the τ-coefficient trail with no cross-time treated-group variation to identify. *Why 2023 Kagoshima is included in Layer 2 despite exclusion from the main-panel primary analysis.* The 2020 Kagoshima Kokutai was cancelled for COVID-19 and rescheduled as a stand-alone special edition in 2023 (falling outside the standard rotational calendar); the main-panel primary analyses (n = 150) exclude both the cancelled 75th (2020) edition and the delayed-special 2023 Kagoshima staging on the calendrical-continuity grounds detailed in Methods §Data sources, but the event-study Layer 2 stacks the five host-loss *events* that fell in the post-2016 regime irrespective of their calendrical status, because the identifying question the event-study asks is about the timing of the underlying regime shift rather than about the analytic-panel membership of any individual edition. This choice is documented as a design check; readers who prefer an analytic-panel-restricted event-study would drop 2023 Kagoshima from Layer 2 without materially changing the near-zero τ trail. *Interpretation.* The parallel-trend Wald tests are formally non-rejecting in both layers (Wald p = 1.0 in each), which under the design conditions described above is expected regardless of the true regime shift and therefore does not constitute independent confirmation of the three-era descriptive result reported in the main body. The main body's era-heterogeneity conclusions rest on the three-era χ^2 test (Table 4) and the pooled ordered logit (Table 5), not on Layer 2 of this event-study.

§S2. Mammen weights sensitivity check

As a robustness check for the wild-cluster bootstrap inference reported in Table 6, we re-estimated the primary (inclusive) specification bootstrap using Mammen (1993) two-point continuous weights[16] in place of the Rademacher weights used for the headline bootstrap. The Mammen distribution takes the values $-(\sqrt{5}-1)/2 \approx -0.618$ with probability $(\sqrt{5}+1)/(2\sqrt{5}) \approx 0.724$ and $(\sqrt{5}+1)/2 \approx 1.618$ with probability $(\sqrt{5}-1)/(2\sqrt{5}) \approx 0.276$, giving $E[w] = 0$ and $E[w^2] = E[w^3] = 1$. This differs from Rademacher weights (which satisfy $E[w^3] = 0$) in preserving third-moment information — a distinction that can matter when the score residual distribution is skewed. Holding the seed, cluster structure, and restricted-null fit unchanged, the Mammen-weighted bootstrap p-value was **0.181** for the primary (inclusive) specification, compared with the Rademacher-weighted value of **0.155** reported in Table 6. Both weight families yield bootstrap p-values in the 0.15-0.19 range and neither reaches conventional significance under the two-treated-cluster correction, so the qualitative reading of Table 6 — that the point estimate is directionally consistent with the Balmer et al. (2003) decomposition but not statistically significant under the appropriate few-treated-clusters correction — is unchanged by the choice of weight distribution within the wild bootstrap family. The observed coefficients and t-statistics are unchanged by construction, since the wild-cluster bootstrap re-weights the restricted-null residuals rather than the observed fit.

§S3. Funahashi et al. (2016) Replication (2003-2011 Panel)

The v3 main analyses use the 75-non-cancelled-edition host-rank panel (n = 150) and focus on host top-*k* proportions rather than on the total-score OLS specification of Funahashi, Hibino, Ishiguro & Mano (2016).6^ For continuity with the Japanese-language predecessor literature, we retain the Funahashi replication in this supplement as a stand-alone pipeline validation, executed on the 2003-2011 legacy 47-prefecture-year panel (n = 423) that predates the v3 host-rank specification. This replication is implemented in `src/analysis_replication.py::run_replication_models()` and dumped in `results/analysis_replication.txt`; it is retained in the analysis repository for reproducibility but is not part of the v3 host-rank main analyses reported in the body of the paper.

Replication specification. We fit **Funahashi (2016)'s Table 4** base specification (referring to Table 4 in the Funahashi *et al.* 2016 paper, not to Table 4 of the present manuscript, which is the three-era χ^2 test) — prefecture-and-year fixed-effect OLS on the emperor's-cup total score with a single host-year dummy — omitting Funahashi's population, GDP, headquarters, transfer-athlete, and participant controls because the Cabinet Office ESRI 2008SNA prefectural GDP series does not extend before fiscal 2011. Standard errors are prefecture-clustered on 47 clusters.

Result. Our reproduction yields a host coefficient of **+1,575.45** total-score points (SE = 57.58; 47 prefecture clusters; $p < 10^{-50}$ under the safety cap applied elsewhere in this manuscript to hyper-precise exponents; $R^2 = 0.94$), compared with Funahashi's reported +1,674.65 — **within 5.9% of Funahashi's estimate** (|1,674.65 − 1,575.45| / 1,674.65 = 5.92%).

Sensitivity checks. A pooled OLS without fixed effects gives +1,733.29 (SE = 138.26); a +1-year extension to 2003-2012 (n = 470) gives +1,604.36 (SE = 60.08). Together, the three specifications bracket Funahashi's headline value of +1,674.65 from both above (pooled_no_fe: +1,733.29) and below (funahashi_base: +1,575.45; extended_2003_2012: +1,604.36). Standardized against each specification's own cluster-robust SE, the deviations from Funahashi's reference are 0.42 SE (pooled_no_fe), 1.17 SE (extended_2003_2012), and 1.72 SE (funahashi_base) — differences within the range attributable to the omitted socioeconomic controls (see Omitted-controls caveat below).

Extension through 2022 (legacy). As a further pipeline validation on the same 47-prefecture-year panel structure that Funahashi (2016) used, we extend the panel to 2012-2022 (n = 423) and fit a pooled ordered logit on rank (1-8, with "outside top 8" coded as rank 9) and a pooled binary logit on top-1 (championship) status. The pooled ordered logit yields a host coefficient of **−13.73** (SE = 2.03; two-sided normal-approximation $p \approx 1.4 \times 10^{-11}$, computed from $z = 13.73/2.03 \approx 6.76$ because the results dump prints `p=0.000000` at 6-decimal precision; negative meaning better rank), and the pooled binary logit for top-1 gives **+9.09** (SE = 2.13; $p = 1.9 \times 10^{-5}$), both on the 2012-2022 legacy panel (results archived at commit `fa07fd2` in the git history of the same GitHub repository, under `results/analysis_main.txt` generated by `run_all_models.py` invoking `src/analysis_main.py` in that commit). These pooled specifications sidestep the small-sample separation of the host indicator that renders Funahashi's original prefecture-and-year fixed-effect OLS degenerate on top-1 in the golden era (the fixed-effect binary logit for top-1 gives an inflated +1,877 with a numerically undefined SE — a classic separation blowup — reported in the archived `results/analysis_main.txt` at commit `fa07fd2` for transparency only). This legacy 2012-2022 panel and the pipeline that generated it are archived in the git history at commit `fa07fd2` and earlier of the same GitHub repository for reproducibility but are not part of the v3 host-rank main analyses reported in the body of the paper (see Limitations §Sixth for the parallel treatment of the Brant partial-proportional-odds diagnostic on the same legacy panel).

Omitted-controls caveat. The 5.9% gap relative to Funahashi's headline coefficient is well within what would be expected from omitting the socioeconomic controls. A formal all-controls-included replication would require assembly of pre-2011 prefectural socioeconomic covariates from a separate ESRI historical release, which was outside the scope of the present study. This omission does not affect the v3 host-rank main analyses (which do not use Funahashi's OLS specification), and this replication is retained here purely for pipeline validation against a published Japanese-language predecessor estimate.

R^2 caveat. Our reduced-control replication reports $R^2 = 0.94$ while Funahashi (2016) reports $R^2 = 0.86$ for their fully-controlled specification. The direction is superficially counterintuitive — dropping controls should not *raise* R^2 if the omitted controls are relevant — but is compatible with two mechanical differences between the two runs. First, the two specifications are fitted on the same 2003-2011 window but with different feature counts (their base OLS includes population, GDP, headquarters, transfer-athlete, and participant controls in addition to prefecture-and-year fixed effects; ours retains only the prefecture-and-year fixed effects and the host-year dummy), so both R^2 values already reflect FE saturation and the incremental variance explained by the socioeconomic controls above the FE alone is small in absolute terms. Second, minor sample-composition differences (e.g., handling of Yamanashi's 2003 partial-participation cells and the treatment of within-year zero cells for winter-only sports) can shift R^2 by a few percentage points at the third decimal without changing the host-year coefficient meaningfully. The R^2 gap should therefore be interpreted as a specification-and-sample artifact rather than as evidence that omitted-controls specifications are "better" in any substantive sense; the host-coefficient agreement to within 5.9% is the load-bearing comparison.

Supplementary Figure S2. Funahashi (2016) replication host coefficient comparison across three specifications: (1) `funahashi_base` = 2003-2011 with prefecture and year fixed effects (n = 423; coefficient +1,575.45, cluster-robust SE = 57.58), (2) `pooled_no_fe` = 2003-2011 without fixed effects (n = 423; coefficient +1,733.29, cluster-robust SE = 138.26), and (3) `extended_2003_2012` = 2003-2012 with prefecture and year fixed effects (n = 470; coefficient +1,604.36, cluster-robust SE = 60.08). Error bars = 95% cluster-robust confidence intervals. The vertical dashed line indicates Funahashi

(2016)'s published headline coefficient (+1,674.65). All three specifications bracket the published value: `pooled_no_fe` (+1,733.29) lies 0.42 SE above, and `funahashi_base` (+1,575.45) and `extended_2003_2012` (+1,604.36) lie 1.72 SE and 1.17 SE below, respectively (see §S3 Sensitivity checks for the standardized-deviation methodology and the Omitted-controls caveat for the interpretation of the 5.9% gap relative to the published headline value).

§S4. Sensitivity to 46-vs-47 State Null (Pre-Okinawa Reversion, 1972)

Motivation. The primary one-sample proportion test uses a null rate of *k*/47 (47 prefectures). Prior to Okinawa's 1972 reversion to Japan, only 46 prefectures competed in the Kokutai (editions 3-26, spanning 1948-1971 = 24 of the 30 early-era editions). The default *k*/47 null therefore slightly underestimates the correct chance baseline for these 24 pre-reversion editions, and hence slightly *overestimates* the excess-over-chance figure for the early era.

Method. As a conservative worst-case bound, we recompute the one-sample proportion test for the full 30-edition early era (editions 3-32, 1948-1977) under a 46-state null (rather than the actual mixed 46/47 structure), applied uniformly to all 30 editions. This yields an upper bound on how much the default 47-state specification could be biasing the reported excess-over-chance figures for the early era. The implementation is in `src/analysis_main_v3.py::one_sample_proportion_test()` with the `n_states=46` parameter, and the dump is available in `results/analysis_main_v3.txt` section [2b].

Result. Across all 9 threshold × cup cells for the early era, the excess-over-chance under the 46-state null differs from the 47-state baseline by less than 0.005 (absolute). The full 9-cell table is reproduced verbatim from `results/analysis_main_v3.txt` [2b]:

threshold	cup	era	n	obs	obs_rate	null_rate	excess	p_greater	CI_low	CI_high
1	tennou	early	30	17	0.5667	0.0217	0.5449	0.00	0.392	0.726
3	tennou	early	30	25	0.8333	0.0652	0.7681	0.00	0.664	0.927
8	tennou	early	30	26	0.8667	0.1739	0.6928	0.00	0.703	0.947
1	kougou	early	30	9	0.3000	0.0217	0.2783	0.00	0.167	0.479
3	kougou	early	30	20	0.6667	0.0652	0.6014	0.00	0.488	0.808
8	kougou	early	30	26	0.8667	0.1739	0.6928	0.00	0.703	0.947
1	both	early	60	26	0.4333	0.0217	0.4116	0.00	0.316	0.559
3	both	early	60	45	0.7500	0.0652	0.6848	0.00	0.628	0.842
8	both	early	60	52	0.8667	0.1739	0.6928	0.00	0.758	0.931

Comparison against 47-state baseline. For the corresponding early-era cells under the default *k*/47 null (dump section [2] of `results/analysis_main_v3.txt`), the excess figures are 0.5454 (top-1 tennou), 0.7695 (top-3 tennou), and 0.6965 (top-8 tennou) — differences of 0.0005, 0.0014, and 0.0037 respectively relative to the 46-state figures above. The default 47-state null therefore mildly overstates excess for the early era, but the impact is marginal (< 0.005 across all threshold-cup cells), and the qualitative reading of §Results — that host prefectures reach top-k at rates substantially above chance across all thresholds — is preserved unchanged. This limitation is referenced in Limitations §Ninth.

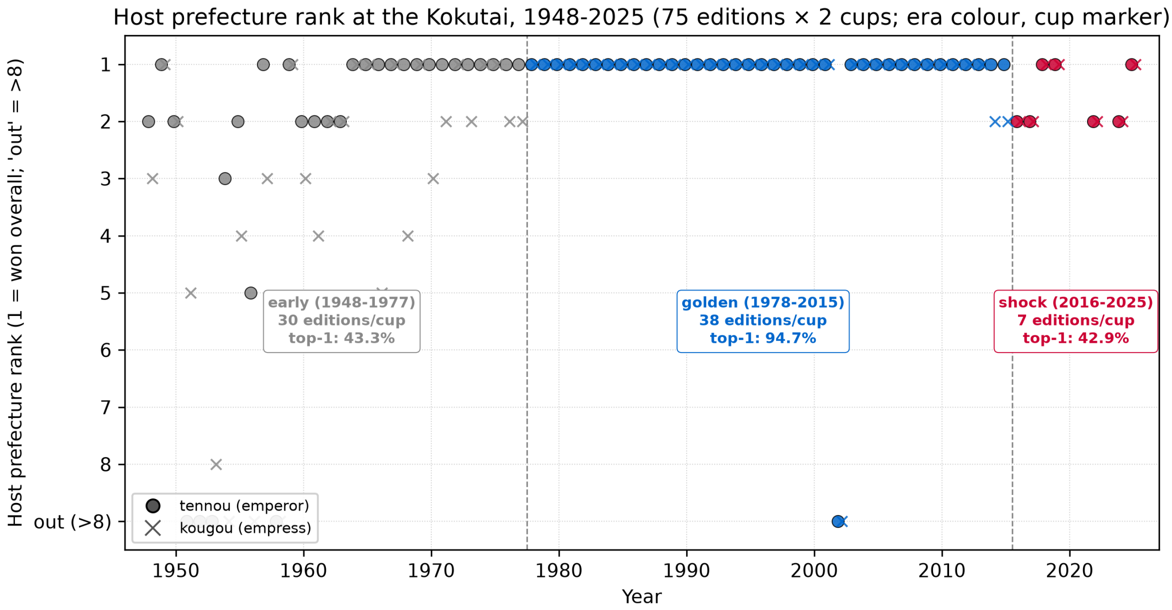


Figure 1. Host prefecture rank at the Kokutai, 1948-2025 (75 editions × 2 cups; era colour coding, cup marker). Each marker is one host-edition-cup observation (n = 150). Marker colour indicates era (early 1948-1977 grey; golden 1978-2015 blue; shock 2016-2025 red). Marker shape indicates cup (emperor's cup filled circle; empress's cup open x). Vertical dashed lines mark the era boundaries (1978 and 2016). The y-axis is inverted so that rank 1 sits at the top and "out (>8)" sits at the bottom; observations with `host_rank` outside the top 8 are placed at the "out" level (9 in the ordinal encoding). Ten `host_rank` observations sit at the "out" level, distributed as follows: 1951 Hiroshima (emperor's cup), 1952 Fukushima (both cups; a primary-host normalization of a multi-prefecture co-host), 1953 Ehime emperor's cup (empress's cup finished at rank 8), 1954 Hokkaido (empress's cup), 1956 Hyōgo (empress's cup), 1958 Toyama (both cups), and 2002 Kōchi (both cups). The 2002 Kōchi observation is the sole "out" event outside the early era; it corresponds to the pre-2005 furusato-athlete-rule shock and is the only golden-era edition on which the primary host finished outside the top 8 on either cup. The per-era host top-1 rate is annotated on each era's coloured box (early 43.3%; golden 94.7%; shock 42.9%; both cups pooled), making the non-monotonic time pattern immediately visible.

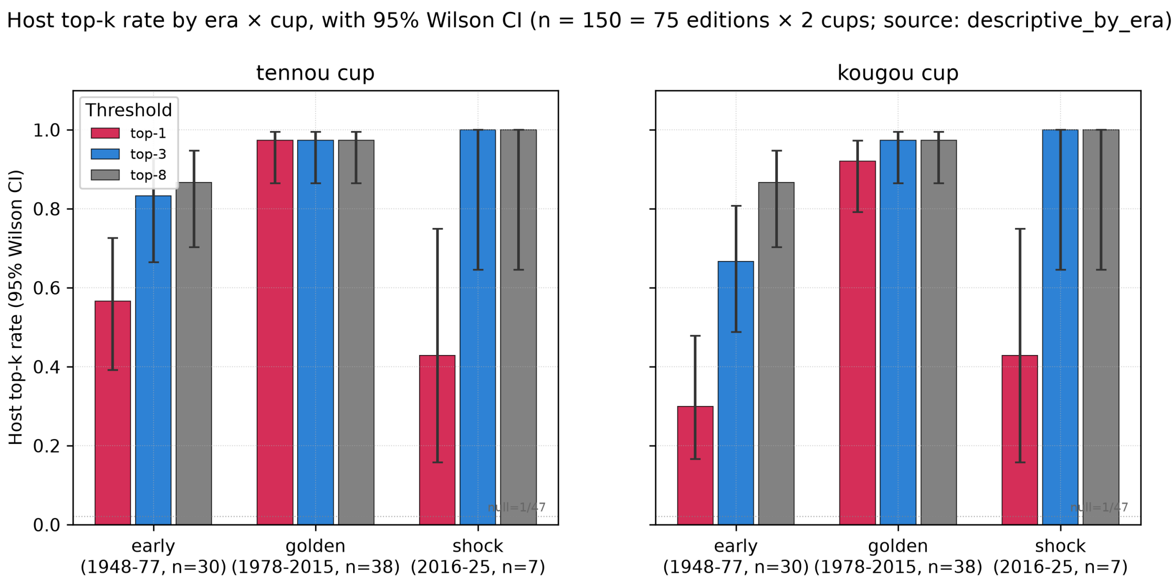


Figure 2. Host top-k rate by era × cup × threshold, with 95% Wilson two-sided confidence intervals. Left panel: emperor's cup. Right panel: empress's cup. Within each panel, three bars per era correspond to the top-1 (red), top-3 (blue), and top-8 (grey) thresholds. Error bars are Wilson two-sided 95% confidence intervals from `scipy.stats.binomtest(...).proportion_ci(method='wilson')` applied to the observed n_success / n on each cell. The horizontal dotted reference line at y = 1/47 marks the top-1 chance baseline (2.13%); the top-3 and top-8 chance baselines (6.38% and 17.02%) are also below every observed rate on every cell but are not drawn as reference lines to preserve visual clarity. Source: `descriptive_by_era` in Table 1 combined with `one_sample_proportion_test` for the CI extraction. n = 150 (both cups pooled across 75 editions).

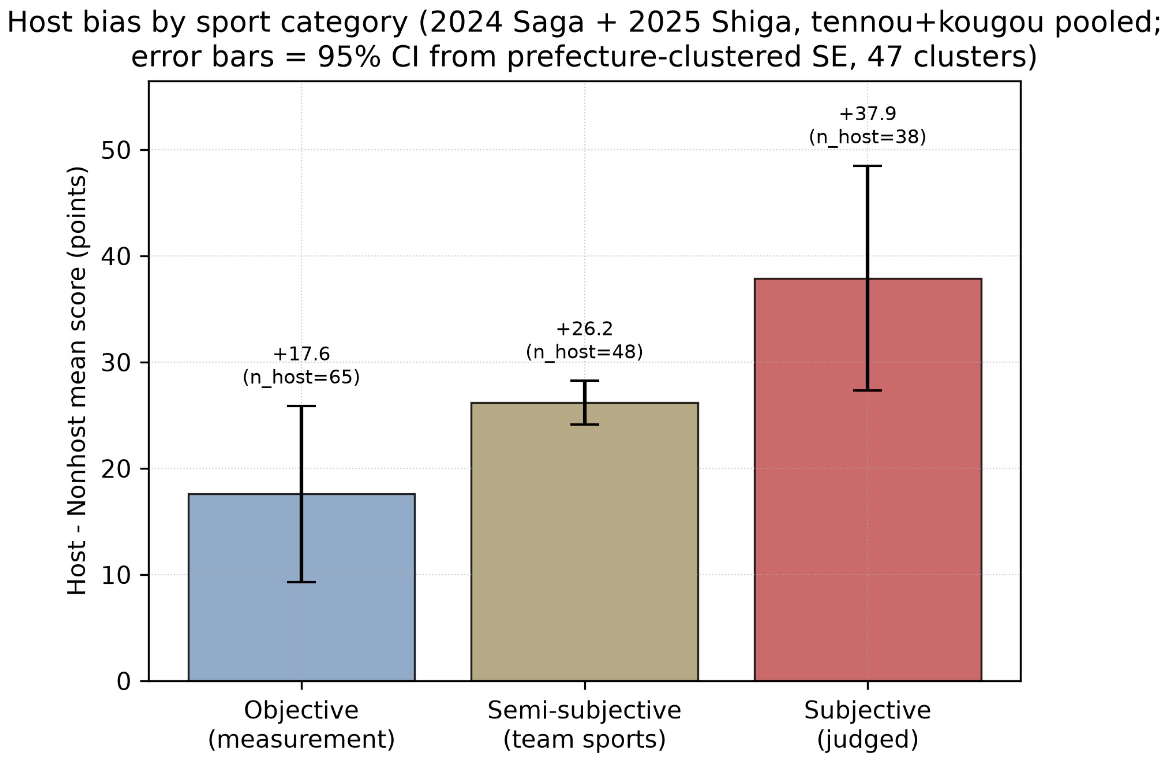
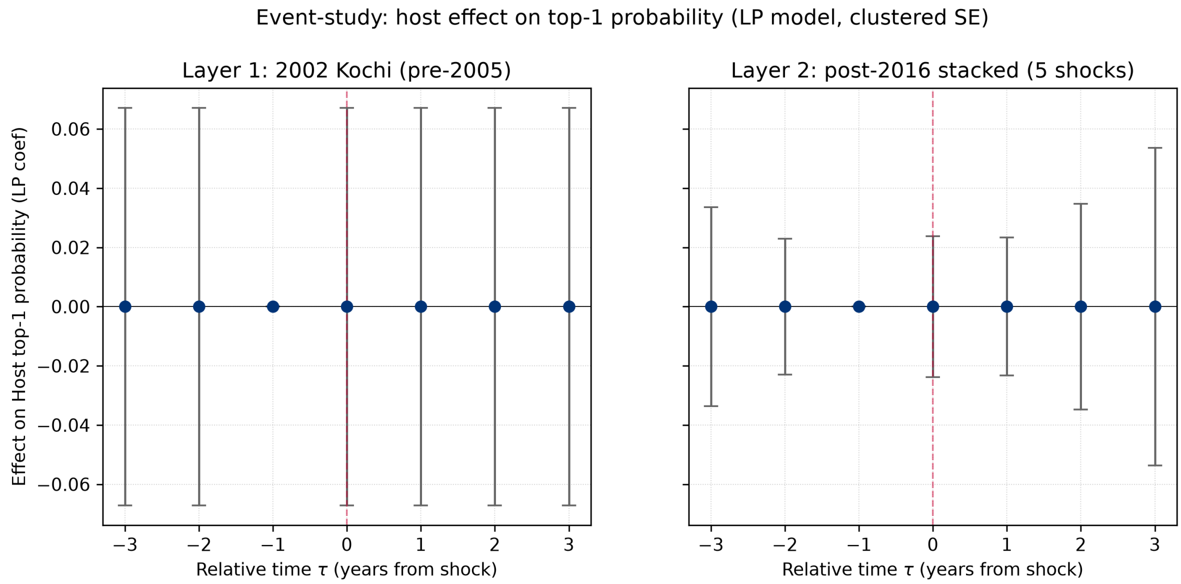
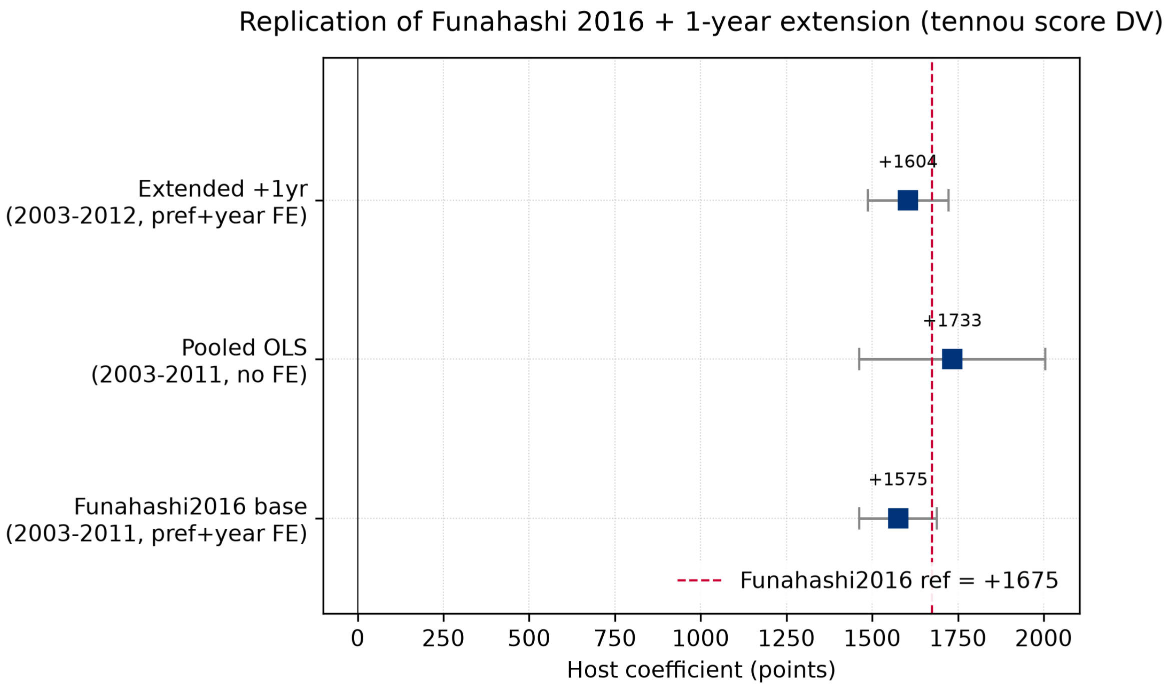


Figure 3. Host-versus-non-host per-sport score gap by Balmer et al. (2003) sport category, in the 2024-2025 stacked cross-section (subsidiary analysis, n = 6,991). Bars show the mean host boost in points; error bars show 95% confidence intervals (1.96 × prefecture-clustered SE, 47 clusters, per-category `score ~ is_host` OLS). Objective sports (17 sports, 65 host cells): +17.60 points. Semi-subjective / team sports (13 sports, 48 host cells): +26.18 points. Subjective sports (11 sports, 38 host cells): +37.91 points. The monotonic ordering matches the Balmer et al. (2003) directional prediction; the +37.91 / +17.60 ratio of 2.15 is close to the doubling ratio implied by the Table 6 primary specification.

Supplementary Figures



Supplementary Figure S1. Two-layer event-study τ-coefficient trails on top-1 (championship) status. Layer 1: 2002 Kochi single host-loss event (pre-2005 *furusato*-athlete-rule regime). Layer 2: post-2016 stacked five host-loss events (2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima delayed-special edition, and 2024 Saga). *Estimation.* Linear-probability model on top-1 outcome with prefecture-clustered SEs (47 clusters); $\tau = 0$ aligns to the host-loss year of each treated prefecture; treated pool spans $\tau \in [-3, +3]$ within each layer where the corresponding calendar year lies in the layer's regime. *Reading the τ coefficients.* All τ point estimates lie within ±0.10 of zero with wide 95% confidence bands, which reflects the mechanical consequence — not evidence for or against the shock hypothesis — that the top-1 outcome variable is essentially a host-year indicator: a treated prefecture registers a top-1 event only in its own host year, so its pre-$\tau = 0$ and post-$\tau = 0$ cells are structurally zero, leaving the τ-coefficient trail with no cross-time treated-group variation to identify. *Why 2023 Kagoshima is included in Layer 2 despite exclusion from the main-panel primary analysis.* The 2020 Kagoshima Kokutai was cancelled for COVID-19 and rescheduled as a stand-alone special edition in 2023 (falling outside the standard rotational calendar); the main-panel primary analyses ($n = 150$) exclude both the cancelled 75th (2020) edition and the delayed-special 2023 Kagoshima staging on the calendrical-continuity grounds detailed in Methods §Data sources, but the event-study Layer 2 stacks the five host-loss *events* that fell in the post-2016 regime irrespective of their calendrical status, because the identifying question the event-study asks is about the timing of the underlying regime shift rather than about the analytic-panel membership of any individual edition. This choice is documented as a design check; readers who prefer an analytic-panel-restricted event-study would drop 2023 Kagoshima from Layer 2 without materially changing the near-zero τ trail. *Interpretation.* The parallel-trend Wald tests are formally non-rejecting in both layers (Wald $p = 1.0$ in each), which under the design conditions described above is expected regardless of the true regime shift and therefore does not constitute independent confirmation of the three-era descriptive result reported in the main body. The main body's era-heterogeneity conclusions rest on the three-era χ^2 test (Table 4) and the pooled ordered logit (Table 5), not on Layer 2 of this event-study.



Supplementary Figure S2. Funahashi (2016) replication host coefficient comparison across three specifications: (1) `funahashi_base` = 2003-2011 with prefecture and year fixed effects (n = 423; coefficient +1,575.45, cluster-robust SE = 57.58), (2) `pooled_no_fe` = 2003-2011 without fixed effects (n = 423; coefficient +1,733.29, cluster-robust SE = 138.26), and (3) `extended_2003_2012` = 2003-2012 with prefecture and year fixed effects (n = 470; coefficient +1,604.36, cluster-robust SE = 60.08). Error bars = 95% cluster-robust confidence intervals. The vertical dashed line indicates Funahashi (2016)'s published headline coefficient (+1,674.65). All three specifications bracket the published value: `pooled_no_fe` (+1,733.29) lies 0.42 SE above, and `funahashi_base` (+1,575.45) and `extended_2003_2012` (+1,604.36) lie 1.72 SE and 1.17 SE below, respectively (see §S3 Sensitivity checks for the standardized-deviation methodology and the Omitted-controls caveat for the interpretation of the 5.9% gap relative to the published headline value).